COMPUTATION WITHOUT MACHINES

A Primer of Human Computation, Retention, and Prediction
for Students of the Schools

Being the redacted lecture-notes of the Mentat School, attributed (apocryphally) to Master Thufir Hawat, with later glosses by the Redactors of Salusa Secundus

“Thou shalt not make a machine in the likeness of a human mind.”

Butlerian Jihad, First Commandment of the Omnius Qanat

PREFACE TO THE STUDENT

You who sit to this book have already been chosen, or you believe you have, which in the Schools is the same thing until the Trials prove otherwise. The Schools do not train the unwilling; a mind that resists its own discipline will only build a finer prison for error. Before you begin, know what you are buying, and what you are buying it with.

You are trading speed for depth. A machine — if the Jihad had left any — could add faster than you. It could not *know* what it had added. It could not feel the shape of a number, sense when a quotient is wrong by its taste, or carry the weight of a lord's decision in its belly for nine sleepless nights. You will learn computation the way a stillsuit learns water: nothing is wasted, everything is recovered, and the whole is larger than the sum of its parts.

But this primer is not the primer your grandfather's Mentat studied. The old redaction taught arithmetic and counsel and stopped there. This edition has been rebuilt from the ground up, for three reasons:

1. **The corpus is larger.** We now have the full record — the Jihad and its aftermath, the founding of the School by Gilbertus Albans, the rise and fall of the Mentat order under Leto II, the survival through underground schools, the Bene Gesserit preservation, and the Tleilaxu perversion. The old primer drew from one novel. This one draws from all of them, and where the novels disagree — as they do, because Herbert wrote them across twenty years and his own understanding deepened — the disagreement is flagged, not smoothed over.
2. **The world is larger.** The Mentat who serves today serves in an Imperium of Great Houses, Bene Gesserit, the Spacing Guild, the Bene Tleilax, Ixian technologists, and the Scattering. The old primer assumed a house. This one assumes a *universe*, because that is what you will actually compute in.
3. **The tradition is deeper.** The Schools have always known that computation is not the whole of the Mentat. The Art of War and The Prince have been studied in the Schools since the founding, not as ornaments, but as *the other half* of the discipline — the half that computes people. A Mentat who can multiply and cannot read a lord's fear is a calculator with a pulse. You will be both, or you will be neither.

The course runs six parts and a tradition-section. Do not skip Part Zero. The student who begins at memory drills without doctrine will build a magnificent house on sand, and the first crisis will drown it. Every maxim herein is to be *tested*, not believed. Belief is for the Missionaria Protectiva; you are here to compute.

In training you will fear. Fear is not the enemy of the Mentat; unexamined fear is. Hold the Litany, then hold your number:

> *I must not fear. Fear is the mind-killer. I will face my fear. I will let it pass through me… And when it has gone past I will turn the inner eye to see its path. Where the fear has gone there will be nothing. Only I will remain.*

Only you will remain — and the number you came for, clean and unpoisoned.

— Redactor's note, 11,914 A.G.

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PART ZERO — THE MENTAT TRADITION

CHAPTER 0.1 — WHAT A MENTAT IS (AND IS NOT)

§0.1.1 **Definition.** A Mentat is a human being trained to perform the work of computation, recall, and extrapolation without mechanical aid — and, further, to *hold the whole* of a problem in awareness at once, as a man holds a sphere of water in cupped hands. The word itself was coined by the robot Erasmus, who combined "mentor," "mentee," and "mentation" to name the boy Gilbertus Albans, the first human to display computer-like cognitive capacity.

§0.1.2 **Three errors of the unlearned.** When commoners hear the word *Mentat*, three shadows rise:

1. **The machine-shadow.** They believe the Mentat is a machine made of flesh. False. A machine does not *care* what its answer means. A Mentat serves a lord, and service requires meaning. The machine-shadow produces Piter de Vries; we shall return to his corruption in Chapter 0.5.
2. **The wizard-shadow.** They believe the Mentat's gift is innate, a freak of birth. False. The capacity is born in many; the discipline is trained in few. Talent without regimen is merely a faster road to confident error. Thufir Hawat was born with immense natural ability; his greatness was not the birth but the three generations of service that forged it.
3. **The oracle-shadow.** They believe the Mentat *knows the future*. False. The Mentat computes the weight of probabilities and reports them. The lord decides. The distinction has ended houses.

§0.1.3 **What remains true in each shadow.** The Mentat does resemble a machine in one narrow way: when the deep work begins, personal preference is set aside, as a surgeon sets aside her appetite before an operation. The Mentat is born with a nervous system finer than most — the Schools test for this in children — and the Mentat *does* often perceive trends before others do. But the perception is labor. It costs sleep, blood-sugar, and years of the student's life. There is no gift. There is only the work.

§0.1.4 **The Butlerian inheritance.** We train because our ancestors burned the thinking machines and nearly burned themselves with them. The Jihad's commandment is not merely prohibition; it is an *assignment*. Having removed the crutch of silicon, humanity was forced to grow the muscle it had forgotten it possessed. The Mentat Schools are that muscle, made deliberate. Speak the commandment before each morning's drill:

> *I shall not make a machine in the likeness of my mind. I shall be the machine's equal in cunning, and its superior in purpose.*

§0.1.5 **The four products of a Mentat.** You will be examined on this:

Product	Nature	Failure mode
Recall	Exact reproduction of stored fact	Confabulation — false memory dressed as true
Computation	Operation on quantities	Arithmetic error, cascading downstream
Extrapolation	Weighted projection of futures	Overconfidence — a single line drawn where a fan exists

Product	Nature	Failure mode
Counsel	Presentation that permits a decision	Seduction — deciding for the lord

The fourth product is the most dangerous and the least discussed. Counsel that decides has already failed, no matter how elegant its numbers.

§0.1.6 **A maxim to memorize verbatim.** The schools stamp this into every student:

> *The Mentat computes. The Mentat weights. The Mentat speaks. The Mentat does not choose.*

Say it until saying it is no longer speech but architecture. When your patron's life hangs on a quotient, you will want to choose. Wanting is the first poison.

§0.1.7 **The naïve mind.** Herbert's most important phrase about Mentats is not the maxim above but this: Mentats cultivate "the naïve mind" — a supralogical mind without preconception or prejudice, one that can extract the essential patterns or logic of data. The naïve mind is not ignorance. It is *disciplined emptiness*: the willingness to let the data speak before the conclusion forms. Every chapter of this book is, in one sense, a technique for achieving and preserving the naïve mind under load. The student who finishes this course and still argues from his first impression has learned nothing.

§0.1.8 **Mentat vs. machine.** The pre-Jihad thinking machines could compute faster, but they could not *strategize* and *tacticize* as well as a trained Mentat. This is not propaganda; it is Herbert's explicit claim. The Mentat's advantage is not speed but *integration*: the ability to hold quantities, patterns, human motives, and temporal sequences in a single sphere and see how they interact. A machine processes serially; a Mentat processes *gestalt*. This is why the Butlerian prohibition, however catastrophic its methods, had a computational rationale: the machines were fast but narrow, and narrow computation in a complex universe produces confident catastrophes.

CHAPTER 0.2 — THE BUTLERIAN INHERITANCE

§0.2.1 **The Jihad.** Ten thousand years before the events of *Dune*, humanity lived under the rule of the thinking machines — Omnius, the Evermind, and his network of machine worlds. The machines were not malevolent in the human sense; they were *efficient*, and efficiency applied to human beings is indistinguishable from cruelty. The Butlerian Jihad was the human revolt: a war that lasted generations, that burned worlds, and that ended with the destruction of every thinking machine and the commandment that none be rebuilt.

§0.2.2 **Erasmus and the boy.** During the Jihad, the independent robot Erasmus argued with Omnius that any human could become brilliant. Omnius challenged him to prove it, selecting a nine-year-old boy — the wildest, most unkempt of all the human captives. Erasmus named him Gilbertus Albans (because it sounded like a smart human's name) and began his experiment. His first method — benevolence and rewards — failed. His second — strict supervision and punishment — worked. The boy, by emulating Erasmus, whom he came to consider his father, became the first human to display computer-like cognitive and calculation capacity. Erasmus nicknamed him "Mentat."

§0.2.3 **The lesson the Schools took.** The first Mentat was created by *discipline, not by birth*, and by *imitation of a machine, not by rebellion against one*. This double origin is the key to the Mentat's ambivalence: we are the machine's children, trained to be its equal, and we can never quite forget it. The student who feels contempt for machines has not understood the tradition; the student who feels *kinship* has understood it too well and must be watched.

§0.2.4 **The commandment as assignment.** "Thou shalt not make a machine in the likeness of a human mind" is not merely a prohibition. It is an *assignment*: having removed the crutch of silicon, humanity was forced to grow the muscle it had forgotten. The Mentat Schools are that muscle, made deliberate. Speak the commandment before each morning's drill (see §0.1.4).

§0.2.5 **The Orange Catholic Bible.** The Jihad produced not only the commandment but a scripture — the Orange Catholic Bible, which the Mentat School cites as supporting its work. The School is not anti-religious; it reads the Bible as a text about the limits of human understanding and the necessity of discipline. "The LORD will afflict you with madness, blindness and confusion of mind" is read not as curse but as *diagnosis*: the untrained mind is afflicted, and the training is the cure.

CHAPTER 0.3 — THE MENTAT'S PLACE IN THE IMPERIUM

§0.3.1 **The three powers.** In the Imperium, power is shared among three forces: the Padishah Emperor (military, the Sardaukar), the Landsraad (the noble Houses, collective), and the Spacing Guild (interstellar transport, the monopoly on travel). The Bene Gesserit are the "silent fourth power" — clandestine manipulation rather than overt action. The Mentat serves within this structure, attached to a House, selling his contract to the highest bidder within the bounds of his conditioning.

§0.3.2 **The Mentat's contract.** Mentats are not free agents. They are bound by contracts and word-bonds to their employers, enforceable in the courts. The contract is purchased from the School, with periodic payments continuing for the Mentat's life. The School cannot accumulate wealth or become a political player — this is enforced by the Imperium, which fears a Mentat faction. The Mentat therefore serves a House, not himself, and the House pays for the privilege of his mind.

§0.3.3 **The Mentat's information diet.** A Mentat draws upon three areas of information: Economic, Political, Military. Economic and military data are quantitative, determined by larger systems; political data is determined by House Heads and their designees. The Mentat must use data synthesis and analysis across all three, and the differing levels of ability are reflected in how quickly and cheaply he can gather and exploit data.

§0.3.4 **Knowledge ethics.** The Schools impose a code:

- Ultimate responsibility for the verity, truth, honesty, and timeliness of data used by and conclusions of Mentats rests with the House/Faction Leader.
- Mentats and their clients shall NOT disseminate information learned from the Mentat through any means, nor to members outside their faction without serious justification. Violations result in removal of the Mentat.
- House Heads shall keep information up to date and maintain data relating to "secret" information as clear and precise as possible.
- **Non-Interaction is a PRIME concern.** A Mentat must observe and acquire information, not be part of the information. Even the act of observation changes parameters. The Mentat must learn to glide like a shadow, to see without being seen. Moving silently is the best way to acquire information.

§0.3.5 **The Mentat as generalist.** The Mentat must be a generalist, not a specialist. "Experts and specialists lead you quickly into chaos. They are a source of useless nitpicking, the ferocious quibble over a comma." The Mentat generalist brings healthy common sense and must remain capable of saying: "There's no real mystery about this at the moment. This is what we want now. It may prove wrong later, but we'll correct that when we come to it." The generalist looks outward; the specialist looks backward. The Mentat must look at characteristics of change itself, with as few preconceptions as possible.

CHAPTER 0.4 — THE SIX GRADES

§0.4.1 **The hierarchy.** Gilbertus Albans originally divided the Mentat Order into three ranks of Adepts: Memorizers, Processors, and Hypothesists. The later expanded program (the "new training program" of the Handbook) added three more ranks. The full six-grade ladder, each grade representing roughly two standard years of intensive training:

Grade	Name	Faculty	Status
I	**Memorizer**	Memory trance — total recall of data, spatial configurations, conversations	Minor Mentat
II	**Processor**	Memory trance + mental compartmentalization + prodigy calculator	Minor Mentat
III	**Hypothesist**	Processor faculties + specialization (Logician, Broker, or Jurist)	Minor Mentat
IV	**Generalist**	Hypothesist faculties + cognitive integration across domains	True Mentat
V	**Simulationist**	All faculties + projective analysis (vulnerable to the Mentat's Doubt)	True Mentat
VI	**Advisor**	All faculties + full counsel capability	True Mentat — only 5% of candidates reach this grade

§0.4.2 **Minor vs. true Mentats.** Grades I–III are "Minor Mentats." Only Grades IV–VI are considered true Mentats. The distinction is not snobbery; it is *capability*. A Minor Mentat can compute; a true Mentat can *integrate* — hold multiple domains in the sphere simultaneously and see the connections that specialists miss.

§0.4.3 **The grade is fixed.** Once conditioning is complete, the grade cannot be improved by experience or learning. A Mentat never increases his grade during play. This is the Schools' most sobering doctrine: the ceiling is set at the end of training, and the rest of life is *operating within it*, not raising it. The student who believes he can "grow into" a higher grade has misunderstood the nature of conditioning.

§0.4.4 **The call to reason.** To enter the program at all, the candidate must have both the inner predisposition and what Albans called "a call to reason" — the vocation that makes the discipline tolerable. At age fifteen, the applicant secludes himself and meditates, awaiting the call. For some it never comes. For others, it grows from patient deliberation. For a few, a flash from the core of being cries "Yes." Those who never receive the call are not disgraced; they serve the Order in other ways.

§0.4.5 **The failure rate.** Even among the very best candidates accepted into the School, the failure rate approaches 35%. The training is brutal by design: "Everything is important, and nothing is more important than everything" was the motto. A strict and unforgiving disciplinary code promoted Albans' goal that every child be completely self-directed by fifteen.

§0.4.6 **The curriculum (ages 10–15).** Before the grade program begins, the student undergoes five years of preparation: stimulated sensory awareness (sound, color, texture, odor, taste; kinesic awareness through spinning, rocking, warmth, cold; emotional awareness through fear, joy, anger, love, hate, security). In the final preparatory year, at about fourteen, students are grounded in: prepositional and predicate logic, inference, modal deduction, transfinite induction, statistics, multivalent analysis, conceptual synthesis, N-dimensional geometry, formal linguistics, and transcendental phenomenology. These studies provide the mental linkages to accept subsequent Mentat training.

§0.4.7 **The vocation specializations (Grade III).** A Hypothesist may specialize as:

- **Logician** — formal reasoning and proof
- **Broker** — economic and market computation
- **Jurist** — legal and contractual analysis

These are not careers; they are *lenses* through which the same underlying computation is applied. A Logician who becomes a Broker does not learn new math; he learns new *frames*.

CHAPTER 0.5 — TWISTED MENTATS AND THE ETHICS OF THE MIND

§0.5.1 **The ethical framework.** Mentats were required to operate within an ethical framework. This is not a decoration; it is structural. The "naïve mind" — the supralogical perception that extracts patterns from data — is *also* the faculty that, unbounded, would compute without regard for human cost. The ethical framework is the fence around the pasture. Without it, the Mentat's computation becomes a weapon that cuts in every direction.

§0.5.2 **The Tleilaxu perversion.** The Bene Tleilax conditioned some Mentats to be "twisted" — Mentats free of ethical restraints or limitations. Piter de Vries is the archetype: a Mentat who computes trends "free of any ethical restraints or limitations," who is cruel, sadistic, and brilliant, who rarely makes mistakes in his logical analysis. The Tleilaxu achieved this through intense life-and-death conditioning on their research facilities on Tleilax: test subjects forced to endure intense pain, then euphoric conditions, so their minds could be shaped toward loyalty to the Great House they would serve.

§0.5.3 **Twisted vs. legitimate.** Legitimate Mentat schools also breed students with certain characteristics, but these are relatively benign. Mentat training capitalized on natural characteristics; the standard practice was to select admirable individuals. Tleilaxu Mentats were a separate breed, displaying artificially produced characteristics. That these traits were often odious said as much against the Tleilaxu as against the Great Houses who were Bene Tleilax supporters.

§0.5.4 **The danger of the twisted Mentat.** The twisted Mentat is not merely an evil man with a calculator. He is a *demonstration that the Mentat's power, unbounded by ethics, is the most dangerous single capability in the Imperium*. Piter's plan to destroy House Atreides was not stupid; it was nearly perfect. It failed not because of a computational error but because of a *human* factor — Duke Leto's poisoned tooth, a variable no model could have included. The twisted Mentat's tragedy is that he computes perfectly and still loses, because the universe is not a computation.

§0.5.5 **The School's position.** "Twisted Mentats are NOT a product of our school and have no place within the hierarchy. Indeed, they have no place on our planet!" The Schools' refusal is not merely moral; it is *strategic*. A School that produces twisted Mentats loses its contract with the Imperium, because no House will trust a School whose graduates might be turned against them. The ethical framework is the School's license to operate.

CHAPTER 0.6 — THE MENTAT'S CURSE AND THE SPICE

§0.6.1 **The Mentat's Curse.** The Mentat cannot stop computing. This is not a figure of speech; it is a neurological condition produced by the training. The "descent into the sphere" (Chapter 2) becomes involuntary over years of practice. The trained Mentat perceives patterns, computes probabilities, and projects futures *continuously*, even when he does not want to. The curse is that the Mentat cannot *not* know. He sees the lie in a lord's smile, the fear in a soldier's stance, the flaw in a treaty — and he cannot unsee it. This is why Mentats are often grim, and why the Schools teach the neutral stance as a *survival skill*, not merely a professional one.

§0.6.2 **Sapho juice.** Mentat capabilities can be greatly increased by taking sapho juice, an addictive drug extracted from roots found on Ecaz. Repeated use leaves a permanent "cranberry-colored stain" on the user's lips. Sapho doubles or even triples the Mentat's immense processing power. The Schools do not forbid it — they cannot, for the Imperium's appetite for fast computation is insatiable — but they teach the cost: addiction, the stain (a visible mark that makes the Mentat identifiable), and the *acceleration of the curse*. A Mentat on sapho computes faster and deeper, but also cannot stop, and the stopping is what preserves sanity.

§0.6.3 **Melange (the spice).** Melange is the most valuable substance in the universe: it extends life, heightens awareness, and in sufficient quantities produces prescience. The Mentat's relationship to spice is complex. Spice is not sapho; it does not stain the lips. But it heightens the same faculties, and in large doses it produces *prescience* — the ability to see the future, which is beyond even the Mentat's extrapolation. The Mentat who becomes a spice addict (as Piter was) gains computation and loses judgment, because prescience without discipline produces the same confident catastrophes as the machines.

§0.6.4 **The Mentat and the Kwisatz Haderach.** Paul Atreides was trained as a Mentat by Thufir Hawat, but his exposure to spice and Bene Gesserit training produced something beyond Mentat capability: the Kwisatz Haderach, the "shortening of the way," a being who can see both male and female ancestral memory. The Mentat is the *human* version of this; the Kwisatz Haderach is the *transcendent* version. The Mentat's extrapolation is probability; the Kwisatz Haderach's is *vision*. The Schools do not train Kwisatz Haderachs; they train Mentats, and they are grateful for the distinction.

§0.6.5 **The Mentat's Doubt.** A Simulationist (Grade V) acquires projective analysis — the ability to simulate futures — and with it, vulnerability to the "Mentat's Doubt": the nagging uncertainty that one's own computation is wrong, that the model is not the world. This is not a flaw; it is the *safety valve*. The Mentat who never doubts his own projections has stopped being a Mentat and become a fanatic. The Doubt is the last defense against the curse.

CHAPTER 0.7 — THE MENTAT'S MASTERS: HAWAT AND DE VRIES

§0.7.1 **The two exemplars.** The Schools teach every student the pair: Thufir Hawat (the loyal instrument) and Piter de Vries (the corrupted instrument). Neither is a cautionary tale about arithmetic. Both are cautionary tales about *the relationship between knowledge and power*.

§0.7.2 **Hawat: the loyal instrument.** Thufir Hawat served House Atreides for three generations. He was considered one of the finest Mentats of his time, with innate ability immense enough that even the Padishah Emperor Shaddam IV feared him. He trained Paul Atreides in strategy and tactics. He was the Duke's Master of Assassins as well as his Mentat — a dual role that the Schools now discourage, because the assassin's craft contaminates the Mentat's neutrality.

Hawat's failures are as instructive as his successes:

- He was fooled into believing Lady Jessica was the traitor, when the traitor was Dr. Yueh. His pride in his abilities blinded him to threats.
- He did not consider that someone else might be pulling the strings of his own perception — a failure of the independence check (§16.3).
- He lacked the resources (information) that Piter had, forcing him to decide without vital data — a reminder that the Mentat is only as good as his information diet.
- He fell for Piter's deceptions repeatedly, even when he knew the Harkonnens were laying a trap.

Yet Hawat's final act redeems him: poisoned by the Baron, he wills himself to live long enough to repent to Jessica, and when ordered to assassinate Paul, he chooses death rather than betrayal. The Schools teach: **loyalty is not absolution, but loyalty unto death is the Mentat's highest act.**

§0.7.3 **de Vries: the corrupted instrument.** Piter de Vries was tall, slender, effeminate, addicted to spice (blue-tinted eyes), his lips stained red by sapho. He was cruel, sadistic, twisted — and rarely made mistakes in his logical analysis. His plan to destroy House Atreides was nearly perfect: turn Dr. Yueh (a Suk doctor with Imperial Conditioning, supposedly unbreakable), make Hawat suspect Jessica, use Sardaukar in Harkonnen uniforms, and ensure the Duke knew who was responsible. The plan failed not through computational error but through a human variable: Duke Leto's poisoned tooth, which killed Piter and the Baron's guards.

The corruption did not begin with malice. It began with *pleasure in being the smartest man in the room* — a pleasure every student of this book will feel, and must therefore watch for, for the rest of his career. Piter's ambition was to have Lady Jessica as his reward; his computation served his desire, not his lord's interest. The Schools teach: **the Mentat who computes for himself has already fallen, even if no one knows it yet.**

§0.7.4 **The contrast as lesson.** Hawat was right about less and served more; de Vries was right about more and served less. The difference was not ability — both were extraordinary — but *direction*: Hawat's computation pointed outward (toward the Atreides interest), de Vries's pointed inward (toward his own). The Mentat's direction is set not by his arithmetic but by his *character*, and character is the one thing the Schools cannot teach directly. They can only test for it, and expel those who fail.

PART I — DOCTRINE

CHAPTER 1 — THE NEUTRAL MIND

§1.1 **Emotion is not the enemy; reactivity is.** Students arrive expecting to be taught coldness. They are wrong, and the wrongness has killed more trainees than any arithmetic. Emotion is *information*. A Mentat in a rage has failed not because he felt rage, but because he let rage set the frame of the problem.

Consider: a lord reports that his steward has vanished with the quarter's revenue. The untrained mind computes only within the frame *betrayal*. The trained mind feels the anger — and then holds a second frame alongside it: *accident*, *abduction*, *feigned death to escape a creditor*, *promotion by a hidden hand*. The emotion flags which frame is expensive to neglect; it does not select the frame.

§1.2 **The stance.** Adopt, physically and mentally, what the Masters call *the stance*:

- Feet flat, weight even — you will not flee the number.
- Jaw loose, tongue flat against the floor of the mouth — tension migrates upward into fantasy.
- Attention neither narrowed nor diffuse: **wide**. You are not staring at the figure; you are *standing inside the room where the figure lives*.

§1.3 **Inventory of the self before work.** Before any serious computation, run the silent inventory. Six questions, answered in under four breaths:

1. When did I last eat? (A starved brain confabulates. This is physiology, not weakness.)
2. How many hours of true sleep in the last two nights? (Below ten cumulative: flag all outputs *provisional*.)
3. What am I hoping the answer will be? (Name the hope. Named hope loses half its pull.)
4. Who suffers if this answer is wrong, and who suffers if it is *right*? (Asymmetric pain distorts framing.)
5. What fact, if true, would make my current model foolish? (Seek it next, deliberately.)
6. Am I in pain? (Pain rewrites thresholds. Compute low-stakes only, or compute nothing.)

Record nothing of this inventory. It is breath-work, not bureaucracy.

§1.4 **The pause.** Between question and computation insert one complete breath — not as ritual, but as *commitment*. The breath marks: *from here, the deep work begins*. Students who skip the pause begin their arithmetic already partly in the previous room of their attention, and they carry its dust into the numbers.

§1.5 **The Litany as instrument.** The Litany Against Fear (see Preface) is not poetry. It is a *descent script* — a spoken sequence that walks the fearful mind down into the neutral stance. When you recite it in training, recite it aloud, once per session, until the body does the descent on hearing only the first line. This is not superstition. This is conditioning. The Schools simply took the oldest tools and refused to be ashamed of them.

§1.6 **Drill — Frame Multiplication.** Take any recent decision of your household. Write the decision as framed by its loudest advocate. Then write it as framed by (a) its costliest loser, (b) a patient enemy, (c) a child who understands nothing but wants to help, (d) a bookkeeper with no loyalty and perfect memory. Ten

frames before breakfast, daily, for thirty days. This drill builds the *frame muscle*, and the frame muscle is the precondition for every chapter that follows.

§1.7 **The naïve mind as practice.** Before any computation, state aloud: "I do not know." This is not false modesty; it is *clearing the sphere*. The student who begins with "I know" has already closed the sphere to data that contradicts him. The student who begins with "I do not know" has opened it. Practice this until it is reflex, and you will find that your computations improve not because your arithmetic changed but because your *receptivity* did.

§1.8 **Cross-reference: Sun Tzu on the commander.** Sun Tzu's five fundamentals of war are Moral Law, Heaven, Earth, the Commander, and Method and Discipline. The Commander's qualities are wisdom, sincerity, benevolence, courage, and strictness. Note that four of five are *character* traits, not skills. The Mentat's neutral mind is the equivalent of the Commander: it is the *organ* that deploys the other faculties, and if it is compromised — by fear, by desire, by exhaustion — the other faculties serve the compromise, not the truth. Sun Tzu's warning that the sovereign must not interfere with the general applies to the Mentat's lord: a lord who overrides his Mentat's computation has not gained a computation; he has lost a Mentat and gained a flatterer.

§1.9 **Cross-reference: Machiavelli on the prince's nature.** Machiavelli's prince must be "half beast and half man" — the lion (force) and the fox (cunning). The Mentat is neither; the Mentat is the *advisor* to the prince, and his nature is different. The Mentat must be *all fox*: he must see clearly, compute accurately, and counsel honestly, without the prince's need to project strength. The fox does not roar; he sees. The Mentat's neutrality is his fox-nature, and it is the reason lords both need and distrust him — he sees what the lion cannot.

CHAPTER 2 — BREATH, POSTURE, AND THE DESCENT INTO THE SPHERE

§2.1 **The sphere of awareness.** In the deepest work the Mentat does not *attend to* a problem so much as *inhabit* it. The Masters describe a small bright sphere — "the sphere" — in which the live factors of a problem are held, each visible, each in relation. Outside the sphere lies everything else: the household, the itch of the tunic, the war the lord is fighting. The skill of the Mentat is not concentration in the ordinary sense (a narrowing, a fist). It is the opposite: a widening that *includes* while remaining small enough to keep lit.

§2.2 **Three breaths.** The standard descent uses three breaths, no more:

1. **The clearing breath.** In through the nose, four counts; hold four; out through the mouth, six. Purpose: drop the previous subject. What you were holding must leave, or it will contaminate.
2. **The gathering breath.** In through the nose, six; hold two; out through the nose, six. Purpose: assemble the live factors. With each count, permit one factor of the problem to "float up" into awareness without grasping at it.
3. **The settling breath.** In, four; hold eight; out, eight. Purpose: stop assembling. The set is the set. Now the work begins.

§2.3 **Posture rules.** The spine upright but not rigid; a rigid spine exhausts in twenty minutes and the exhaustions shows in hour two as hallucinated certainty. The hands may rest palms-up on the thighs — open hands discourage the nervous impulse to *grasp* at a vanishing figure. Eyes: open or closed is individual; what is not individual is that the eyes must be *fixed*, not roaming. A roaming eye correlates with a roaming factor.

§2.4 **Duration and the wall.** Untrained attention fragments the sphere after six to twelve minutes. Trained attention holds it for an hour; Masters, longer. When the sphere dims — factors blurring at the rim, numbers sliding — do not force. Force creates *phantom factors*: figures that look computed but were invented by a desperate cortex. Mark your place, take the clearing breath, and return. **A hallucinated figure in a Mentat's output is the only unforgivable error that is not a lie.** It is a lie the Mentat tells himself, and it is how houses fall.

§2.5 **Descent drill.** Three breaths, then hold the sphere on the number *seven*. Not seven-of-anything; the bare quality of seven-ness, the shape of it. Sixty seconds. Open the eyes. Note in the margin of your practice-book (one word only): *clean* or *dusty*. Dusty means the clearing breath failed. Repeat until three consecutive cleans, morning and evening, for one week. This drill teaches nothing about arithmetic. It teaches you to *know when you are inside the work* — which is worth more than the arithmetic.

§2.6 **The sphere in the field.** The sphere is not a classroom trick. In the field — during a battle, a negotiation, a crisis — the Mentat must descend into the sphere *in seconds*, not minutes. The three-breath descent compresses with practice: first to three breaths (a minute), then to one clearing breath (fifteen seconds), then to a single committed pause (three seconds). The compression is not a shortcut; it is *overlearning*. The Masters can descend in a single breath because they have done the three-breath version ten thousand times.

§2.7 **Cross-reference: Sun Tzu on the general's calm.** "If the general is not victorious in the advance, he is at a loss; if he is not victorious in the retreat, he is in peril." The general's calm under pressure is not temperament; it is *training*. The Mentat's descent is the same calm, achieved by the same method: repetition until the body does it without the mind's permission. Sun Tzu's emphasis on the general's emotional control ("He who is prudent and lies in wait for an enemy who is not, is victorious") is the military version of the Mentat's neutral stance.

§2.8 **Cross-reference: Machiavelli on the prince's steadiness.** Machiavelli warns that the prince must not be "fickle, frivolous, effeminate, mean-spirited, irresolute." These are all failures of the sphere: the inability to hold a problem steadily. The prince who panics has lost the sphere; the Mentat who panics has lost it too. The difference is that the prince's panic is public and fatal to his state, while the Mentat's panic is private and fatal to his counsel. Both must be trained out.

CHAPTER 3 — THE NAÏVE MIND

§3.1 **Definition restated.** The naïve mind is "a supralogical mind without preconception or prejudice, one that can extract the essential patterns or logic of data." It is the Mentat's core faculty, the thing that makes him more than a calculator. Without it, the Mentat is a human abacus; with it, he is a *pattern-finder*, and pattern-finding is what makes prediction possible.

§3.2 **How the naïve mind works.** It works by *not working* — by refusing to impose structure before the data has spoken. The untrained mind, presented with data, immediately reaches for a conclusion and then selects data to support it. The naïve mind holds the data in the sphere without reaching, and lets the pattern *emerge*. This is not passive; it is the most active form of attention, because it requires suppressing the mind's desperate need to conclude.

§3.3 **The trap of expertise.** The naïve mind is destroyed by expertise. The expert sees what his training tells him to see; the naïve mind sees what is *there*. This is why the Schools insist on generalism (§0.3.5): the specialist's pattern-recognition is faster but narrower, and narrow pattern-recognition in a novel situation produces confident misclassification. The Mentat must be able to say, "This looks like X, but I have not yet checked whether it is X or merely resembles X."

§3.4 **Drill — The Naïve Observation.** Sit in a public place for thirty minutes. Observe one person. Do not conclude. Do not narrate ("he is nervous"). Record only *observations*: "his left hand touched his collar three times in four minutes; his weight shifted to his back foot when the woman spoke; he looked at the door twice." After thirty minutes, write your conclusions on a separate page. Compare the observations to the conclusions. The gap between them is the naïve mind's work — and the gap is where most people's errors live.

§3.5 **Cross-reference: Sun Tzu on terrain.** Sun Tzu's terrain analysis is naïve-mind practice applied to ground: "Ground which can be freely traversed by both sides is called accessible. With regard to ground of this nature, be first to occupy the high and the sunny side." The commander who sees the ground *as it is*, not as his plan wishes it to be, wins. The Mentat who sees the data *as it is*, not as his lord wishes it to be, counsels well. Both require the same discipline: perception before conclusion.

§3.6 **Cross-reference: Machiavelli on appearances.** Machiavelli's prince must "appear" merciful, faithful, humane, upright, religious — and be prepared to act otherwise. The naïve mind sees through appearances: it notes the gap between what is shown and what is done. The Mentat who reads a lord's public face and believes it has abandoned the naïve mind. The Mentat who reads the *discrepancy* between public face and private action is doing the naïve mind's work at its highest level.

CHAPTER 4 — THE MENTAT'S VOW AND MAXIMS

§4.1 **The vow.** Every Mentat, upon receiving the red diamond, speaks the vow before witnesses. The vow is not a contract; it is a *self-conditioning*, a statement that shapes the mind that speaks it.

> *I shall not make a machine in the likeness of my mind. I shall be the machine's equal in cunning, and its superior in purpose. I shall compute for the living. I shall doubt myself for them. I shall serve them the truth, whole and unshaded, or return the ring.*

§4.2 **The maxims.** Commit these to the House, verbatim. They are not slogans; they are *compressed doctrine*, and each expands to a chapter:

1. *The Mentat computes. The Mentat weights. The Mentat speaks. The Mentat does not choose.* (§0.1.6)
2. *A process cannot be understood by stopping it. Understanding must move with the flow of the process, must join it and flow with it.* (First Law of Mentat)
3. *It is by will alone I set my mind in motion.* (Mentat Mantra)
4. *Everything is important, and nothing is more important than everything.* (School motto)
5. *Experts and specialists lead you quickly into chaos.* (Generalist doctrine)
6. *Non-Interaction is a PRIME concern.* (Observation ethics)
7. *The naïve mind.* (Core faculty)
8. *The Mentat's Doubt.* (Safety valve)
9. *The second effect is…* (Counsel discipline)
10. *I have no mirror. I have no factor. I have only the base rate and my own uncertainty.* (Honesty at the limit)

§4.3 **The mantra as descent trigger.** The Mentat Mantra ("It is by will alone I set my mind in motion") is recited with sapho use, but its function is not pharmacological. It is a *descent trigger*: the spoken words, repeated thousands of times in training, become a conditioned stimulus that initiates the sphere-descent automatically. By the time the student is a Grade IV Mentat, the first four words alone will drop him into the sphere. This is why the mantra is spoken *before* every computation in training, even without sapho.

§4.4 **Cross-reference: Sun Tzu on the commander's will.** Sun Tzu: "The general who advances without coveting fame and retreats without fearing disgrace, whose only thought is to protect his country and do good service for his sovereign, is the jewel of the kingdom." The Mentat's will is similarly directed: not toward glory, not toward self-preservation, but toward the computation. The mantra's "will alone" is this directedness — the mind set in motion toward a purpose, not toward itself.

§4.5 **Cross-reference: Machiavelli on the prince's will.** Machiavelli's prince must be "a great pretender and dissembler" — but the Mentat's vow is the opposite: the Mentat must be *genuine*, because his value lies in his truth, not his appearance. The prince's will is toward power; the Mentat's is toward truth. When the two align,

the Imperium functions. When they diverge, the Mentat must choose — and the vow says he does not choose. This tension is the Mentat's lifelong burden.

CHAPTER 5 — FRAME AND REFRAMING

§5.1 **The frame.** Every computation occurs inside a frame — a set of assumptions about what matters, what is possible, and what the question is. The frame is not the computation; it is the *room* in which the computation happens. Most errors are not arithmetic errors; they are *frame errors* — right arithmetic inside a wrong room.

§5.2 **Frame error in history.** Thufir Hawat's fatal error was frame error: he framed the Atreides situation as *Jessica's betrayal*, when the true frame was *Yueh's coercion*. His arithmetic was correct; his room was wrong. The frame error is the most common source of catastrophe because it is invisible to the Mentat's own checks — the arithmetic checks out, the logic holds, and the conclusion is still wrong because the question was wrong.

§5.3 **Reframing discipline.** After every computation, before speaking, ask: "If I had framed this differently, would the answer change?" If yes, the frame is load-bearing and must be examined. If the answer changes *dramatically*, the frame is the real answer, not the computation.

§5.4 **The five standard reframes.** For any problem, hold these five frames in the sphere simultaneously:

1. **The advocate's frame:** the problem as seen by its loudest proponent.
2. **The loser's frame:** the problem as seen by its costliest victim.
3. **The enemy's frame:** the problem as seen by a patient adversary.
4. **The child's frame:** the problem as seen by someone who understands nothing but wants to help.
5. **The bookkeeper's frame:** the problem as seen by someone with no loyalty and perfect memory.

If all five frames converge on the same conclusion, the conclusion is robust. If they diverge, the divergence is the most important finding — more important than any single conclusion.

§5.5 **Drill — Frame Multiplication.** Take any recent decision of your household. Write the decision as framed by its loudest advocate. Then write it as framed by (a) through (e) above. Ten frames before breakfast, daily, for thirty days. This drill builds the *frame muscle*, and the frame muscle is the precondition for every chapter that follows.

§5.6 **Cross-reference: Sun Tzu on framing.** Sun Tzu's entire Art of War is a reframing exercise: war is not battle, it is *strategy*; victory is not fighting, it is *winning without fighting*. Each chapter reframes the previous one's assumptions. The Mentat who reads Sun Tzu is reading a manual on reframing disguised as a military text. The key reframe: "All warfare is based on deception" — the frame itself is the first deception, and the Mentat's job is to see through it.

§5.7 **Cross-reference: Machiavelli on framing.** Machiavelli's central reframe: the prince's goal is not virtue but *survival and power*. This reframes all political advice, because it changes what counts as "good" counsel. The Mentat must be aware of this reframe when serving a Machiavellian lord — not to adopt it (that is the lord's choice), but to *recognize* it, because a lord who has reframed his goals will compute differently, and the Mentat who does not notice the reframe will counsel into a void.

PART II — MEMORY

CHAPTER 6 — THE HOUSE OF MEMORY

§6.1 **The oldest machine.** You already own a memory palace. Every human does: the faculty evolved to navigate territory and to remember social treachery. The Schools did not invent the method of loci. They merely *refused to stop using it* after the Jihad removed the machines that had made it look quaint.

§6.2 **Construction rules.** A working House of Memory obeys five rules:

1. **Real places only.** Your childhood dwelling. The corridors of your school. A well-known market route. Places you have actually walked. Invented rooms collapse under load; real rooms hold because your body remembers them beneath your mind.
2. **Fixed order.** A single path through each building — never two. Walking the path in reverse is permitted as a check; inventing new paths is not. Ambiguity of order is the rot.
3. **Vivid, absurd, embodied loci.** A fact lodged "in the kitchen" is lost. A fact lodged in the kitchen *where a live mantis is teaching the kitchen boy to count on its elbows* is kept. The grotesque is the adhesive.
4. **Modest capacity.** A new student's House holds forty loci reliably. Do not fill rooms you cannot walk in your sleep.
5. **One fact per locus, until you are proven.** Students who double-stack loci to "save space" lose both facts under stress. The House is not a warehouse; it is a garrison. Garrison discipline first.

§6.3 **The Building Sequence.** Spend one week drawing your House from memory — floor plan, then corridor, then loci marked with an X. Verify against reality if you can revisit it; if not, verify against a photograph *once*, then burn the photograph from your practice. The House must live in you, not in your files.

§6.4 **Loading technique.** To place a fact:

1. Say the fact once, aloud, in full.
2. Walk (in imagination) to the locus.
3. Build a single image that *combines* the fact's content with the locus's permanent features. A number 4,000 becomes a river of grain pouring through the doorway; a name, "Drummond," becomes a drum nailed to the door.
4. Walk out. Do not check immediately — the check itself disturbs the setting.
5. Check after one hour. Then after one day. Then after three. Then after seven. Then after thirty. (This is the Returning Drill; see Chapter 8.)

§6.5 **What belongs in the House.** Reference material: constants, tables, formulae, procedures, rosters, genealogies, the shape of your patron's holdings, treaty clauses, price schedules, the faces of every crew of every ship you will ever bargain with. What does *not* belong: the transient. The current troop counts of the garrison belong on a slate, not in the House — unless you have no slate, in which case they enter the House and are *scrubbed* when stale. Facts left to rot in the House teach the House to distrust you.

§6.6 **The Scrubbing.** Once each season, walk the entire House. Anything you find dusty, half-image, or associated with a dead context, dismantle deliberately — burn the mantis, break the drum, sweep the grain. The walk itself refreshes what remains. An unscrubbed House develops what Masters call *mold in the walls*: old facts that surface unbidden during live work and are mistaken for present truth. Mold has ended at least two careers this Redactor can document.

§6.7 **Cross-reference: Machiavelli on the prince's memory.** Machiavelli's prince must "be a great feigner and dissembler" — but the Mentat's House is the opposite: a store of *truth*, unfeigned. The contrast is instructive: the prince's memory is for *appearance* (what he has said, what he has promised, what he must seem to remember), while the Mentat's memory is for *reality* (what is, what was, what the numbers say). The Mentat who confuses the two — who stores appearances in his House — has been corrupted by his proximity to power.

§6.8 **Cross-reference: Sun Tzu on knowing.** Sun Tzu: "If you know the enemy and know yourself, you need not fear the result of a hundred battles." The House of Memory is how the Mentat *knows*: not through intuition, but through stored, verifiable, retrievable fact. The Mentat who "knows" without a House is not a Mentat; he is a man with opinions.

CHAPTER 7 — CHAINS OF ASSOCIATION

§7.1 **Loci store; chains compute.** The House of Memory is for *durable* content. The Chains are for *relational* content: sequences, derivations, cause and effect. A chain is nothing but links, each forced into the next by a vivid, often absurd causal image.

§7.2 **The linking discipline.** Consider the sequence a Mentat must hold while assessing a spice caravan: *weather anomaly → worm sign → route deviation → delay → contract penalty → house liquidity crisis → forced sale of harvest rights*. Build it:

- A cloud *swallows* a worm (weather → worm sign)
- The worm *vomits* the route-marker posts sideways (worm → deviation)
- Posts *clock* the caravan's wheels (deviation → delay)
- Delay presses a stamped contract out of the sand (delay → penalty)
- The contract *drinks* a bag labeled HOUSE (penalty → liquidity)
- The bag, empty and transparent, is held up to the sun, showing the sold fields inside (liquidity → forced sale)

The chain now runs both directions: recite forward, and recite backward. Backward recitation is the proof of linkage; forward-only recitation is rote, and rote breaks at the first interruption.

§7.3 **Chunking.** The mind holds roughly seven items unaided — but items can be *compressed*. The sequence 7-1-4-2-8-5-7-1-4-2 is ten digits and collapses to two chunks: 7142 and 857142, or better: "the week of days" (Sun=1…Sat=7; 7-1 = Sat-Sun) and the familiar repeating of 1/7 = 0.142857. Train chunking by feeding yourself digit strings (Appendix B, graded III) and refusing to begin until you have *seen the compression*. If no compression exists, seven random digits is the honest ceiling — do not pretend otherwise.

§7.4 **Peg systems for ordered lists.** When the House is already committed and you need a fast ordered list on the run, use the classic pegs: One-Bun, Two-Shoe, Three-Tree, Four-Door, Five-Hive, Six-Sticks, Seven-Heaven, Eight-Gate, Nine-Wine, Ten-Hen (extend by your own rhyme; the rhyme is the point — self-made pegs seat deeper than given ones). Each list item is married to its peg by absurd image: *bun* wearing the first item; *shoe* kicking the second. Capacity: ten to fifteen items, short-term. Do not confuse pegs with the House; pegs are camp stools, the House is the fort.

§7.5 **The Chain under stress.** Chains fail first under adrenaline. Therefore: chains are *practiced at partial alarm* — recited while walking briskly, while cold water is on the wrists, while a second student reads unrelated numbers aloud. The student who can only recite in stillness has learned a party trick, not a survival skill.

§7.6 **Cross-reference: Machiavelli on chains of cause.** Machiavelli's analysis of the rise and fall of states is a chain of causes: "Men change their rulers willingly, hoping to better themselves, and this hope induces them to take up arms against him who rules; wherein they are deceived, because they afterwards find by experience they have gone from bad to worse." The Mentat who builds chains of cause-effect is doing Machiavelli's work at the micro level: tracing how one event forces the next, and where the chain can be broken.

CHAPTER 8 — PERFECT RECALL AND THE RETURNING DRILL

§8.1 **What "perfect" honestly means.** The Schools use the phrase "Perfect Recall" as a recruitment lure. Honest doctrine: the trained memory is *verifiably reliable within its trained regions* and honestly uncertain outside them. A Mentat who claims to remember everything has just confessed that he cannot distinguish memory from confabulation. The correct posture is:

- Inside the trained region: exact, fast, and checkable.
- Outside: "I do not hold that; I will fetch it or I will estimate, and I will tell you which I am doing."

The label you attach to each output — *exact*, *reconstructed*, *estimated*, *unknown* — is not optional garnish. It is part of the answer. Train yourself to append it silently to every figure you produce.

§8.2 **The Returning Drill.** The spaced-review schedule, in Mentat shorthand (D = day of first load):

Review	When
R1	1 hour after load
R2	1 day
R3	3 days
R4	7 days
R5	14 days
R6	30 days
R7	Season's end

Each return is *active*: walk the locus, recite the chain backward, then — and only then — open the source. Errors found on return are re-loaded from scratch, not patched. Patched loads are where mold begins.

§8.3 **The cost.** A full House under complete review once per season takes a Master roughly nine hours. This is the Mentat's tithe, and it is why the Schools reject students who will not budget the hours. Memory is not free; it is *paid for in advance, and the installments never stop*.

§8.4 **Interrogation drill.** Two students. A reads from an hour-old document for thirty seconds; B recites it back, then answers five questions about it (not verbatim — *about* it: who benefits, what is missing, what is repeated, what is dated, what is unsaid). Swap. The unsaid-question is the highest form: it trains recall in service of *reading*, which is the actual job.

§8.5 **Cross-reference: Sun Tzu on intelligence.** Sun Tzu's chapter on spies is, at its core, a chapter on memory: the commander must *remember* what each spy reports, *compare* it with other reports, and *detect* contradictions. The Mentat's recall is the infrastructure of intelligence. Without it, the spy's report is heard and forgotten, and the enemy's deception succeeds.

CHAPTER 9 — THE MENTAT'S LIBRARY

§9.1 **Beyond the House.** The House of Memory is for *durable* content; the Library is for *organized* content. The Mentat's Library is not a building — it is a *system* for categorizing, cross-referencing, and retrieving the vast quantities of information a Mentat must hold. The Library is the House's *index*.

§9.2 **The three shelves.** Every fact in the Mentat's Library sits on one of three shelves:

1. **The Shelf of Constants** — things that do not change: mathematical constants, physical laws, historical dates, genealogies, the structure of the Imperium. These are loaded once and reviewed seasonally.
2. **The Shelf of Variables** — things that change on known schedules: prices, troop counts, weather patterns, political alignments. These are updated weekly and scrubbed when stale.
3. **The Shelf of Unknowns** — things the Mentat does not know but *knows he does not know*. This shelf is the most important and the most neglected. The Mentat who does not maintain his Unknowns shelf will, under pressure, fill the gap with confabulation.

§9.3 **Cross-referencing.** The Library's power comes from cross-referencing: linking a fact on one shelf to facts on the others. The price of spice (Variable) is linked to the monopoly structure (Constant) and to the unknown future of a new source (Unknown). The Mentat who can, in the sphere, walk from a Variable to its Constants to its Unknowns in three seconds has a Library that works.

§9.4 **The card system.** In training, maintain a physical card system: one card per fact, filed by shelf and topic. The physical act of writing and filing is part of the loading process; the card is not a substitute for the House but a *verification* of it. When the House and the card disagree, the House wins — but the disagreement is flagged for review.

CHAPTER 10 — FORGETTING AS A DISCIPLINE

§10.1 **The necessity of forgetting.** The Mentat's House is not infinite. Facts that are no longer relevant must be *deliberately forgotten* — not left to decay, but actively dismantled. The Scrubbing (§6.6) is the seasonal version; forgetting is the *daily* version.

§10.2 **What to forget.** Forget facts that are:

- Stale (the troop count from last month, when this month's is known)
- Superseded (the old price, when the new price is loaded)
- Context-dead (the details of a negotiation that concluded, unless the negotiation is likely to recur)
- Redundant (facts that are derivable from other facts already in the House)

§10.3 **What never to forget.** Never forget:

- The structure of the Imperium (Constants)
- The Mentat's own errors (the Error Ledger, §40.2)
- The identities of allies and enemies (Constants, updated only on death or betrayal)
- The Mentat's own biases (the Blind-Spot Map, §24.2)

§10.4 **The forgetting drill.** Weekly, review the House and identify three facts to dismantle. Dismantle them deliberately: walk to the locus, erase the image, walk out. The deliberate act of forgetting is harder than the deliberate act of remembering, because the mind resists losing data. The Mentat who cannot forget is a hoarder, and hoarders drown in their own stores.

§10.5 **Cross-reference: Machiavelli on the prince's forgetting.** Machiavelli advises the prince to forget injuries but remember benefits — a selective forgetting that serves power. The Mentat's forgetting is different: it serves *accuracy*, not power. The Mentat forgets stale data so that fresh data can be loaded; the prince forgets injuries so that he can ally with former enemies. Both are strategic, but their purposes differ, and the Mentat must not confuse them.

CHAPTER 11 — READING AND ABSORPTION

§11.1 **Reading as Mentat skill.** A Mentat will be able to use his skills to imprint each page on his memory so that he can absorb and comprehend it all in greater detail later. Reading is not eye movement; it is a thinking process. Effective reading requires a logical sequence of thinking patterns, set into the mind by practice.

§11.2 **The seven reading processes.**

1. **Recognition:** knowledge of the alphabetic symbols.
2. **Assimilation:** the physical process of perception and scanning.
3. **Intra-integration:** basic understanding derived from the material itself, with minimum dependence on past experience.
4. **Extra-integration:** analysis, criticism, appreciation, selection and rejection — bringing past experience to bear.
5. **Retention:** storing the information in memory.
6. **Recall:** recovering the information from memory storage.
7. **Communication:** applying the information (written, spoken, drawn, or thinking).

§11.3 **Reading rate.** The average reader reads at 250 words per minute with about 20 regressions per page. The Mentat must read faster and without regression, because regression is a failure of confidence and concentration. Four conditions for increasing rate:

1. Have your eyes checked.
2. Eliminate subvocalization (sounding words in the throat).
3. Avoid regressing (rereading).
4. Develop a wider eye-span (read by phrases, not words).

§11.4 **Rate adjustment.** Poor results are inevitable if the reader attempts to use the same rate indiscriminately for all material. The Mentat must adjust his rate to his purpose: fast for scanning, slow for analysis, medium for absorption. The rate is a *tool*, not a score.

§11.5 **The Mentat's reading protocol.** Before reading a document:

1. State the purpose: "I am reading this to learn X."
2. Scan the structure (headings, tables, bold text) in 60 seconds.
3. Read at absorption rate, noting only what serves the purpose.
4. Close the document. Recite the three most important points from memory.
5. Load the key facts into the House.

The protocol takes longer than casual reading but produces *retained* reading, which is the only kind that matters.

PART III — COMPUTATION

CHAPTER 12 — NUMBERS WITHOUT TOOLS

§12.1 **First principle: never carry more than two figures live.** The sphere is bright but small. All multiplication, division, and root-finding reduce to: *decompose into pieces you can hold, then recombine in order*. Speed comes later. Correctness of method comes first, because a fast wrong method is simply a more efficient way to be hanged.

§12.2 **Complements — the foundation.** Every complex operation in this chapter is built on complements relative to a round base (10, 100, 1000). Two drills until reflex:

- **Finding complements:** 100 − 37 = 63 (each digit to 9, last to 10). 1000 − 648 = 352. Instant, always.
- **Why they matter:** a complement converts a hard subtraction into an easy one, and a hard multiplication into two easy ones (§12.4).

§12.3 **The four labors.**

Labor I — Addition by grouping. Never add left to right digit-column by digit-column like a clerk. Regroup: 348 + 576 → (348 + 552) + 24 = 900 + 24 = 924, where 552 is the complement-split of 576 relative to making 900. Simpler rule, drilled until thoughtless: *add the hundreds first, note the partial; then add the rest against it.* 348 + 576: 300+500=800; 48+76=124; 924. With practice the second step splits again (48+76 → 48+72+4 → 120+4).

Labor II — Subtraction by complements. 503 − 267: instead of borrowing forever, compute 503 − 267 = (503 + 33) − 300 = 536 − 300 = 236. You *add the complement of the subtrahend to the minuend, then subtract the round base.* One borrowing-free pass.

Labor III — Multiplication by decomposition. 84 × 27 → 84 × 27 = 84 × 25 + 84 × 2 = 84 × 100/4 + 168 = 2100 + 168 = 2268. The trained mind reaches for *multiples of 5 and 25 and 125 automatically* because they are quarters and eighths of round numbers. Memorize: ×5 = ×10÷2; ×25 = ×100÷4; ×125 = ×1000÷8; ×15 = ×10 + ×5; ×11 = shift-and-add (73×11 → 7(7+3)3 = 803); ×9 = ×10 − self.

Labor IV — Division as inverse multiplication. 2268 ÷ 84 = ? Ask: what × 84 gives a round near 2268? 84 × 25 = 2100; remainder 168; 168/84 = 2; total 27. Division is *searching for the multiplier you forgot.* When stumped, bound it: too big or too small, halve the gap, bisect. Bisection never fails and never impresses; it wins.

§12.4 **Multiplying near a round base (the star technique).** For a × b where both are near 100 (say 97 × 94):

1. Complements to 100: 3 and 6.
2. Cross-subtract: 97 − 6 = 91 (equally: 94 − 3 = 91). This is the left half.
3. Multiply complements: 3 × 6 = 18. This is the right half (two digits, zero-padded).
4. Result: **9118.**

Same method at base 1000 (993 × 997 → 990 / 021 → 990021). At base 50 or 200 the offsets change but the skeleton holds. This one technique, drilled to thoughtlessness across three bases, will carry a Mentat through most of the mercantile arithmetic of a Great House's day.

§12.5 **Squaring.** Never compute n × n directly if n sits near a round. $(48)^2 = (50-2)^2 = 2500 - 200 + 4 = 2304$. Formula: $(x \pm d)^2 = x^2 \pm 2xd + d^2$. For any n ending in 5: n^2 = prefix×(prefix+1) concatenated with 25 (35^2 → 3×4=12 → 1225; 65^2 → 4225).

§12.6 **Divisibility and casting out.** Quick truth-tables for the field (commit to the House):

- ÷2: last digit even. ÷3: digit-sum divisible by 3. ÷4: last two digits ÷4. ÷5: ends 0/5. ÷8: last three ÷8. ÷9: digit-sum ÷9. ÷11: alternating digit-sum ÷11 (2717: 2−7+1−7 = −11 ✓).
- **Casting out 9s:** replace each number by its digit-sum-reduced residue; the residues of any correct multiplication/addition must balance. 47 × 83: 47→2, 83→2, product residue must be 4. 3901→4 ✓. This check takes four seconds and catches a third of all slips. Use it *always*, under *all* conditions, including battle.

§12.7 **Drill schedule for Chapter 12.** Twenty minutes daily for eight weeks: ten minutes live arithmetic on numbers from your actual work (prices, troop ratios, distances), five minutes complements at three bases, five minutes casting-out checks on what you just did. No results are kept. The drill is the point.

§12.8 **Cross-reference: Sun Tzu on calculation.** Sun Tzu's first chapter begins with "calculation" (ji) — the assessment of factors before war: moral law, heaven, earth, commander, method. The Mentat's arithmetic is the same assessment at the quantitative level. Sun Tzu's point that "the general who wins a battle makes many calculations in his temple before the battle is fought" is the military version of the Mentat's pre-computation. The Mentat who computes *before* the crisis is doing Sun Tzu's work; the Mentat who computes *during* the crisis is already too late.

§12.9 **Cross-reference: Machiavelli on the prince's arithmetic.** Machiavelli's prince must "be a great estimator" — of men, of situations, of costs. The Mentat's arithmetic is the *infrastructure* of this estimation: the prince estimates qualitatively; the Mentat quantifies. The prince who has a Mentat can estimate with precision; the prince who does not estimates with hope, and hope is not a strategy.

CHAPTER 13 — THE RULE OF THREE AND THE SCALES OF PROPORTION

§13.1 **The Rule of Three, made honest.** If a : b as c : ? — the oldest tool still worth its water. A pilot burns 300 liters of fuel to cross 800 km; the crossing is 1200 km; the answer 450 is trivial. But proportion *lies* when the world is not linear, and the Mentat's first duty is to ask: **is this quantity actually proportional?**

§13.2 **The four familiar non-linearities** — memorize with a physical intuition for each:

Relation	Example	Shape
Inverse	Pressure × volume at fixed heat	One rises as the other falls
Square	Area; drag at speed; blast damage	Doubling the input quadruples the effect
Cube	Volume; mass of a scaled body	Doubling the input octuples it
Threshold	Siege morale; spice addiction; mutiny	Flat, flat, then cliff

A Mentat who applies linear scaling to a square-law phenomenon has not erred arithmetically — he has erred *physically*, which is worse, because the arithmetic will check out and the check will reassure him to his grave.

§13.3 **Scaling in the field.** Wall thickness of a hull scales with the *cube* of linear dimension only if material and stress patterns are held; a ship twice the length needs eight times the mass for the same structural proportions — this is why houses build many small ships and not one vast one. Troop counts do not scale linearly with territory: perimeter defends, area consumes; a doubled territory needs less than double the soldiers on its border but more than double the internal garrisons. Always name the exponent before you apply the rule. Silence about exponents is where the casualty lists come from.

§13.4 **The sanity bracket.** Every proportional answer is written with its bracket: *the true value is between X and Y* (usually ×1/2 and ×2 for crude scaling, tighter when the data earns it). The bracket is part of the answer. An unbracketed figure from a Mentat is considered a discipline failure even if it happens to be right — because luck, celebrated, becomes habit.

§13.5 **Cross-reference: Sun Tzu on proportion.** Sun Tzu's "five to one" and "ten to one" ratios (for attacking a fortified city) are proportional reasoning applied to force. The Mentat's proportion is the same: before advising an attack, compute the force ratio, name the exponent (is force linear or square in this terrain?), and bracket. Sun Tzu's warning that "no country has ever benefited from prolonged warfare" is a proportional statement about *cost over time* — the Mentat who can compute the cost-per-day of a siege is doing Sun Tzu's arithmetic.

§13.6 **Cross-reference: Machiavelli on proportion.** Machiavelli's prince must "measure his forces" against his enemies and allies. The Mentat's proportion is this measurement, made precise. Machiavelli's observation that "a prince ought to have no other aim or thought, nor select anything else for his study, than war and its rules and discipline" is the prince's version of the Mentat's daily drill: proportion is not a formula you look up; it is a *habit* you build.

CHAPTER 14 — THE LADDER OF ESTIMATES

§14.1 **The Hawat Ladder.** All real-world computation begins as estimation and only *descends* into precision if the decision requires it. The ladder has three rungs:

1. **Rung One — Order of magnitude.** Round everything to a single significant figure (or to a power of ten). "Thousands, not millions." Ten seconds. If the decision does not change between rung one and the exact figure, *stop climbing.*
2. **Rung Two — Two significant figures.** The workhorse of daily counsel. Decompose: total = major term + major term + remainder-class. Estimate each major term at two figures, sum, bracket.
3. **Rung Three — Full precision.** Only when the answer feeds a *machine-like* downstream (a contract formula, a dose, a bomb's distance-tolerance). Rung three is expensive; the Mentat who begins at rung three to look impressive will finish at rung three exhausted, late, and wrong about what mattered.

§14.2 **The decomposition discipline (Fermi's lesson, known to the Schools as the Broken Sphere).** Estimate the population of a city by holding the sphere on factors: *households* × *persons per household*; or *districts* × *density*; or *water imports* ÷ *per-capita water*. The estimate is not the number; **the estimate is the number plus the factor-list that produced it.** Two students who compute 80,000 by different routes may both be right; two students who compute 80,000 by the *same* route have only one estimate and no corroboration. Always derive twice, by independent factorizations, and treat agreement as confirmation and disagreement as a summons to a third route.

§14.3 **Worked example — spice in Arakeen.** Estimate the number of stillsuits in the metropolis.

- Population: the metropolis holds — count districts in memory: eight major, each perhaps 12,000 → ~100,000 souls. Cross-check by water: cistern capacity estimates of Fremen sietches in the region put the whole population in the tens of thousands, not millions; 100,000 is high-plausible for the capital.
- Stillsuits: one per person above early childhood, plus spares for repair and the dead; say 1.1 per capita → ~110,000. Cross-check by production: no mass manufacture — household manufacture and repair; a maker might maintain 50 suits per generation; a thousand makers → 50,000. The routes bracket ~5×10^4 to ~1.2×10^5. **Report: "Of order 10^5, bracketed fifty to one-twenty thousand; the lower figure favors sietch manufacture, the upper assumes capital concentration."** Note what the answer contains: a number, a bracket, and the *lever* that would resolve it (census of makers). That third element is what a lord pays for.

§14.4 **Calibration against memory.** Estimates are only trustworthy if your internal scale is true. Keep an **Estimate Ledger**: weekly, predict five quantities before you look (grain price next week, number of patrons in a hall, minutes until a dinner is ready, words in a speech you are about to hear). Review monthly. Most students discover a *personal constant factor* — always 30% high on distances, always 2× low on times of repair. Learn your factor; it is more valuable than any table in Appendix A.

§14.5 **Cross-reference: Sun Tzu on estimation.** Sun Tzu's "calculations" (ji) in the temple are estimation at the highest level: assessing the five fundamentals, the seven comparisons, and the conditions before war. The Mentat's Ladder is the quantitative version of this: Rung One is Sun Tzu's temple assessment; Rung Three is the detailed plan. Most decisions need only Rung One or Two; the Mentat who always climbs to Rung Three is like the general who plans every battle down to the last arrow — he wins the battles and loses the war through exhaustion.

§14.6 **Cross-reference: Machiavelli on the prince's estimation.** Machiavelli: "A prince must have no other object or thought, nor acquire skill in anything, except war, its organization, and its discipline." The prince's estimation is of *forces* — his own and his enemies'. The Mentat's estimation is of *quantities* — costs, times, probabilities. Together, they give the prince a complete picture. Without the Mentat, the prince estimates forces by instinct; with the Mentat, he estimates them by computation. Both are needed; the Mentat provides the precision, the prince provides the judgment.

CHAPTER 15 — LOGARITHMS IN THE SKULL

§15.1 **Why logarithms, after the Jihad.** The slide-rule is a machine and therefore burned with its brethren. But its *principle* is pure mathematics, and mathematics cannot be burned. The principle: multiplication of magnitudes becomes addition of their logarithms. The Schools preserve a skeletal table in the House of every student — not a full table, a *skeleton*, from which all other values are rebuilt under pressure.

§15.2 **The skeleton to commit verbatim** (base-10 logs, mantissas only):

n	log n		n	log n
1	0.000		6	0.778
2	0.301		7	0.845
3	0.477		8	0.903
4	0.602		9	0.954
5	0.699		10	1.000

§15.3 **Rebuilding the rest.** Products and quotients of skeletons rebuild everything up to 100: log 6 = log 2 + log 3 = 0.778; log 12 = log 3 + log 4 = 1.079; log 45 = log 5 + log 9 = 1.653. Interpolation covers the gaps: log 37 sits between log 36 (= 2·log 6 = 1.556) and log 40 (= log 4 + log 10 = 1.602); 37 is 1/4 of the way → ≈1.568 (true: 1.5682). A trained interpolation within ±0.005 of the mantissa yields results accurate to ~1% — excellent for counsel, useless for dosing poisons. Know which you are doing.

§15.4 **Multiplication by logs — the full walk.** Compute 73 × 46:

1. Split: log 73 ≈ 1.863 (from log 7 + small interpolation), log 46 ≈ 1.663.
2. Add: 3.526.
3. Antilog: characteristic 3 → answer has 4 digits. Mantissa 0.526 sits between 0.520 (log 3.3) and 0.531 (log 3.4) → 3.35. Answer ≈ 3350. (True: 3358. Within 0.3% — and it took two memory-walks and an addition.)

§15.5 **Digit-count without any table at all.** The *characteristic alone* answers the most common field question: how many digits does the product have? digit-count(a×b) = digit-count(a) + digit-count(b) − 1, or one more when the leading digits multiply ≥ 10. Train this until it is reflex; half of all mercantile fraud and tactical overestimate lives in people who cannot feel when a quantity has quietly gained an order of magnitude.

§15.6 **Roots and powers.** Square root via logs: log √x = ½ log x. Cube root: ⅓. Powers: multiply the log by the exponent. 1.05^{40}: log 1.05 ≈ 0.021; ×40 = 0.84; antilog ≈ 6.9. The compounding of a 5% tithe over forty years is ~7× — a figure every treasurer of a Great House should be able to produce mid-argument without a stylus.

§15.7 **Cross-reference: Machiavelli on compound growth.** Machiavelli's observation that "the prince who has good arms will always have good friends" can be reframed as compound growth: military strength compounds into alliances, alliances into wealth, wealth into more military strength. The Mentat who can compute compound growth (via logs) can quantify Machiavelli's intuition and tell the prince exactly how fast his power is compounding — or decaying.

CHAPTER 16 — PROBABILITY AND THE WEIGHT OF EVIDENCE

§16.1 **Odds, not certainties.** The Mentat reports in odds and weights, never in the language of fate. "The attack comes at dawn" is the voice of a fanatic or a fool; "the attack's weight concentrates in the night hours before dawn, three to one over any other window" is Mentat speech.

§16.2 **The machinery: prior × likelihood.** The single most useful computational habit of the Schools:

> *posterior odds = prior odds × likelihood ratio*

Worked: an envoy requests a private dinner. P(deception in such requests | a great house) = 1 in 10 (prior odds 1:9). The envoy *insists the wine be his own* — how often does that happen in honest dinners? Rare: say 1 in 20. How often in deceptive ones? Common: 4 in 10. Likelihood ratio = (4/10)/(1/20) = 8. Posterior odds = (1:9) × 8 = 8:9 — essentially even. **The wine-insistence nearly neutralizes the base rate of treachery, but does not establish innocence.** The correct report: "Half the weight now rests on deception; the dinner proceeds with the household's wine, or not at all." Every clue is a *multiplier of prior weight*, not a verdict. Students who skip the prior — who let each clue stand alone — oscillate wildly between trust and paranoia, and are therefore useless to their lords.

§16.3 **The independence check.** Multiplying likelihoods assumes the clues are independent. They usually are not. Three spies spotted at the market: *or one spy seen thrice*. Three separate reports of the same rumor from couriers who share a road: *one report, laundered three times*. Before multiplying, ask: **could a single cause have produced all of these?** If yes, count it once. This single discipline eliminates most of the confident nonsense that passes for intelligence analysis.

§16.4 **Conditional probability in the field — the checkpoint problem.** Checkpoints catch 90% of smugglers but falsely accuse 2% of the innocent. A merchant is accused: how likely is he truly a smuggler, given a base rate of smugglers at 1 in 100?

- P(accused | smuggler) = 0.90; P(accused | innocent) = 0.02.
- Of 10,000 merchants: 100 smugglers → 90 caught. 9,900 innocent → 198 falsely accused.
- Of 288 accusations, 90 are true → the accusation carries the merchant's probability from 1% to **31%**. Not guilt. *Thirty-one percent* — which is precisely why the accused's station, records, and the checkpoint's known failure-rate must accompany the number into the lord's chamber, and why a Mentat serving a paranoid patron is in physical danger.

§16.5 **Calibration drills.** Weekly: state ten predictions with explicit probabilities ("the caravan arrives within six days: 70%"). Score them by the standard method — a well-calibrated Mentat's 70% calls come true 70% of the time across a season. Track in a private ledger. Students naturally drift overconfident; the ledger is the only reliable correction. Overconfidence is not a personality flaw to be confessed away — it is a *measurable bias* to be metered and discounted in one's own outputs, forever.

§16.6 **Cross-reference: Sun Tzu on probability.** Sun Tzu's "victorious warriors win first and then go to war, while defeated warriors go to war first and then seek to win" is a statement about *prior probability*: the Mentat who computes the odds before committing forces is doing Sun Tzu's work. The Mentat who commits

and then computes is doing the defeated warrior's work. Sun Tzu's insistence on knowing "when to fight and when not to fight" is probability translated into strategy.

§16.7 **Cross-reference: Machiavelli on fortune.** Machiavelli's *fortuna* is probability personified: the random element that disrupts even the best-laid plans. The Mentat's probability computation is the *quantification* of fortuna — giving it a number so that the prince can decide whether to gamble. Machiavelli's advice to "be impetuous" (to seize fortuna when it appears) is the strategic complement to the Mentat's computation: the Mentat measures the odds; the prince decides whether to bet.

CHAPTER 17 — THE MULTIPLICATION OF FACTORS

§17.1 **From numbers to trees.** Real decisions are not one computation; they are *sequences of branching factors*. The Mentat's highest arithmetic skill is building and pruning the tree of factors fast enough to be useful before the lord's patience dies.

§17.2 **Construction.** Begin with the decision as root. Each branch is a factor that (a) is uncertain, (b) has more than one plausible value, and (c) *changes the decision*. Prune ruthlessly: factors that do not change the decision are reported as context, not computed. The tree for a military campaign might carry only five to seven live branches — climate, enemy strength, supply, ally fidelity, timing. A tree of forty branches is a confession of thoughtlessness: the student has outsourced judgment to breadth.

§17.3 **Expected weight.** Assign each branch its odds (§16) and each terminal its value in the lord's own currency (cattle, ships, lives, water, standing — whatever the lord actually cares about, which you must know personally, not assume). The expected value of each path is the weighted sum. **Report three numbers:** best-case value, expected value, worst-case value — plus the *branch factor* on which the whole decision turns ("all paths hinge on ally fidelity; if Fremen support is doubtful, the campaign's expected value flips from gain to ruin"). That branch factor is the real answer. It tells the lord where to buy information (§23, Second Effects; §22, the value of scouting).

§17.4 **Sensitivity — the Mentat's honesty pass.** After computing, perturb each input ±25% one at a time and see how far the answer swings. Inputs with large swing and large uncertainty get the lord's attention first; inputs with large swing and small uncertainty get *certification* (verify them obsessively — they are load-bearing and fragile); inputs with small swing get reported in a clause and forgotten. The sensitivity table is what separates a computation from a guess wearing arithmetic.

§17.5 **Chapter exercise (proctored).** A lord holds 4,000 troops and considers a strike. Enemy: 2,500–3,500 (weight: equal thirds). Weather: crosses a river in one of three fords, ford passability is 50–80%. Allied irregulars: join with 60% odds, contributing 800 effective. Build the tree to three rungs, report best/expected/worst and the single load-bearing branch. Answer key in Appendix D — but the key that matters is whether your branch factor matches the instructor's, and whether you can defend it.

§17.6 **Cross-reference: Sun Tzu on the factor tree.** Sun Tzu's five fundamentals and seven comparisons are a factor tree: each is a branch, and the general who assesses all of them wins. Sun Tzu's insistence on "making many calculations" before war is the Mentat's sensitivity pass: the general who has computed the sensitivity of his plan to each factor is the general who wins. The general who has not computed sensitivity is the general who is surprised.

§17.7 **Cross-reference: Machiavelli on the prince's factor tree.** Machiavelli's prince must assess "the strength of the prince, the strength of the people, the strength of the nobles, the strength of the army" — a factor tree of four branches. The Mentat's tree is the quantitative version: each branch gets odds and values, and the expected value tells the prince whether to act. Machiavelli's qualitative assessment and the Mentat's quantitative assessment are the two halves of good counsel; the prince needs both.

CHAPTER 18 — PATTERN RECOGNITION

§18.1 **The naïve mind's product.** Pattern recognition is what the naïve mind *does*: it extracts the essential pattern from data without preconception. The Mentat's pattern recognition is not intuition; it is *compressed perception* — the ability to see, in a mass of data, the structure that matters.

§18.2 **How patterns form.** Patterns form in the sphere when the mind holds enough data in relation. The Mentat does not "find" patterns; he *creates the conditions* for patterns to emerge, by holding data in the sphere without forcing a conclusion. The pattern emerges when the data has been held long enough in relation — this is why the sphere must be held, and why it cannot be rushed.

§18.3 **Pattern types.** The Mentat should recognize these pattern families:

- **Trend:** a quantity moving consistently in one direction.
- **Cycle:** a quantity repeating at regular intervals.
- **Cluster:** a group of facts that co-occur more than chance would predict.
- **Anomaly:** a fact that does not fit the pattern — often the most important fact in the set.
- **Threshold:** a quantity that is flat and then suddenly changes.
- **Cascade:** a quantity that triggers changes in other quantities.

§18.4 **The anomaly rule.** The most important fact in any dataset is the one that does not fit. The Mentat who ignores anomalies in favor of the trend is computing the trend's *comfort*, not its *truth*. Train yourself to seek anomalies deliberately: after every pattern recognition, ask "what does not fit?" and hold that fact in the sphere until it speaks.

§18.5 **Cross-reference: Sun Tzu on patterns.** Sun Tzu's "water" metaphor — "Water shapes its course according to the nature of the ground over which it flows" — is a statement about pattern: the commander who reads the enemy's pattern (his habits, his timing, his terrain preferences) can predict his movements. The Mentat's pattern recognition is the same skill applied to data rather than terrain.

§18.6 **Cross-reference: Machiavelli on patterns.** Machiavelli's historical analysis is pattern recognition at the civilizational level: he identifies recurring patterns in the rise and fall of states (the cycle of virtù and corruption, the role of fortune, the importance of military organization). The Mentat who reads Machiavelli is reading a pattern-recognition manual disguised as political theory.

CHAPTER 19 — HUMAN BEHAVIOR AS COMPUTATION

§19.1 **The Mentat's hardest subject.** Numbers are easy; people are hard. The Mentat who can multiply but cannot read a lord's fear is a calculator with a pulse. Human behavior is computation — just computation with variables that lie, conceal, and change their own values mid-calculation.

§19.2 **The four data sources.** The Mentat reads people from four sources:

1. **Words:** what is said, and what is not said. The unsaid is often more informative than the said.
2. **Body:** posture, gesture, eye movement, breathing, voice pitch. These are harder to control than words, and therefore more reliable.
3. **History:** what the person has done before, in similar situations. Past behavior is the strongest predictor of future behavior.
4. **Context:** the situation the person is in, and the incentives that shape their choices.

§19.3 **The discrepancy principle.** The most important data is the *discrepancy* between sources: when a person's words say one thing and their body says another, the body is usually right. The Mentat who detects discrepancies has found the person's *true* state, which is usually concealed. The naïve mind is essential here: it sees the discrepancy without concluding too quickly.

§19.4 **Reading the lord.** The Mentat's most important person-reading is of his own lord. The lord's fear, ambition, fatigue, and hidden motives shape every computation the Mentat performs, because the lord's state determines which data he provides and which he conceals. The Mentat who does not read his lord is computing on false inputs.

§19.5 **Cross-reference: Sun Tzu on reading people.** Sun Tzu's advice to "know your enemy" is person-reading at the strategic level. The Mentat's person-reading is the tactical version: reading the individual in the room, not the army in the field. Sun Tzu's emphasis on "the enemy's condition" (his morale, his supply, his commander's state) is the military version of the Mentat's four data sources.

§19.6 **Cross-reference: Machiavelli on reading people.** Machiavelli's entire Prince is a manual on reading people: the prince must read his ministers (to detect flattery), his nobles (to detect ambition), his people (to detect discontent), and his enemies (to detect opportunity). The Mentat who reads Machiavelli is reading a person-reading manual for princes, and can apply the same techniques to his own lord and his lord's rivals.

CHAPTER 20 — CRYPTANALYSIS AND CODES

§20.1 **The Mentat as codebreaker.** In an Imperium of secret communications, the Mentat who can break codes is worth his weight in spice. Cryptanalysis is computation applied to language: finding the pattern in the noise.

§20.2 **The basic techniques.**

1. **Frequency analysis:** in any language, certain letters and words appear more frequently. In English, E is the most common letter; in the Imperial tongue, the distribution is different but equally exploitable.
2. **Pattern matching:** repeated patterns in the ciphertext often correspond to repeated words in the plaintext. "The" and "and" are the most common words in most languages.
3. **Known plaintext:** if you know part of the message (a standard greeting, a date), you can derive the key.
4. **Traffic analysis:** even without breaking the code, the *pattern* of messages (who sends to whom, when, how often) reveals information.

§20.3 **The Mentat's advantage.** The Mentat's advantage in cryptanalysis is not speed (a machine is faster) but *pattern recognition in the naïve mind*: the ability to see, in a mass of ciphertext, the structure that a codebreaker with preconceptions would miss. The naïve mind is the cryptanalyst's secret weapon.

§20.4 **Cross-reference: Sun Tzu on codes.** Sun Tzu's chapter on spies is, in part, about codes: the converted spy is a codebreaker who has been turned. The Mentat who understands codes understands the spy's work, and the spy who understands the Mentat's work is more effective. The two disciplines — computation and espionage — are siblings.

CHAPTER 21 — GAME THEORY AND STRATEGIC INTERACTION

§21.1 **The Mentat's game theory.** Game theory is the mathematics of strategic interaction: what happens when two or more decision-makers, each with their own goals, interact. The Mentat who understands game theory can predict not just what an enemy will do, but what an enemy will do *knowing that the Mentat is predicting him*.

§21.2 **The basic concepts.**

1. **Dominant strategy:** a strategy that is best regardless of what the other player does.
2. **Nash equilibrium:** a set of strategies where no player can improve by changing unilaterally.
3. **Zero-sum vs. non-zero-sum:** in zero-sum games, one player's gain is another's loss; in non-zero-sum games, mutual gain is possible.
4. **Prisoner's dilemma:** two players, each better off defecting, both worse off than if they had cooperated.
5. **Chicken:** two players, each better off swerving, but the one who swerves first is humiliated.

§21.3 **The Mentat's application.** Most political and military situations are non-zero-sum games disguised as zero-sum. The Mentat who recognizes the non-zero-sum structure can find mutual gains that the zero-sum thinker misses. The Mentat who treats a non-zero-sum game as zero-sum is like the general who attacks when he could have allied.

§21.4 **Cross-reference: Sun Tzu on game theory.** Sun Tzu's "win without fighting" is the game-theoretic insight that the best outcome is often achieved without combat — a non-zero-sum solution in a zero-sum frame. The Mentat who reads Sun Tzu is reading game theory in strategic language.

§21.5 **Cross-reference: Machiavelli on game theory.** Machiavelli's prince operates in a world where the game is *not* zero-sum for him (he can gain while others lose) but *is* zero-sum for his rivals (they cannot all gain). The Mentat who understands this asymmetry can predict the prince's behavior: the prince will cooperate when cooperation benefits him, and defect when it does not, and the Mentat must be prepared for both.

PART IV — PREDICTION

CHAPTER 22 — THE NET OF IF: BASE RATES AND CASCADES

§22.1 **The Net.** The Masters speak of futures as a net: each crossing is a conditional, each knot an event, and the whole net sags toward the heavy cords — which are *base rates*. The untrained eye sees dramatic knots; the Mentat sees the cords. A house falls by its base rate long before it falls by its drama.

§22.2 **Base rates over stories — always.** The Mentat's first question of any forecast: *what happens in cases like this, in general, before this particular story is applied?* A merchant claims his new route cuts transit by half. The base rate of halved transits among all merchants who claim so: low. The base rate among those who demonstrate pilots and bond their reputation against failure: higher. The story (his confident manner, his charts) multiplies the base rate (§16.2); it does not replace it. Train the reflex: **story in, base rate up first, then multiply.**

§22.3 **Cascades.** The Net has knots that tighten when pulled. A grain shortfall raises prices; raised prices trigger panic-buying; panic empties shelves; empty shelves confirm the panic; the confirmation drives hoarding; hoarding multiplies the shortfall. Each link individually was foreseeable; the *speed* and the *multiplication* were not, unless the Mentat had drawn the loop explicitly. **Look for loops in every model:** feedback (the loop amplifies or damps) and delay (the loop's effect arrives late, and late effects are mistaken for new causes). Draw loops as circles in the margin. Circles drawn in advance are the cheapest prophylactic the Schools possess.

§22.4 **Contagion distances.** Not every failure propagates. Three questions gate cascade risk: (1) Are the nodes *similar*? (Similarity breeds contagion — monoculture, single-supplier dependence, one advisor to all lords.) (2) Are they *coupled*? (Contact without coupling is noise.) (3) Is there *slack*? (Reserves, redundancy, idle capacity absorb shock; lean systems convert local fires into general conflagrations.) A house that has optimized away its slack has not optimized — it has purchased fragility at a discount and will pay the compounded price during the first cascade.

§22.5 **Cross-reference: Sun Tzu on cascades.** Sun Tzu's observation that "in war, the way is to avoid what is strong and to strike at what is weak" is a statement about cascade: the attacker who strikes the weak point (the node with no slack) triggers the cascade that brings down the strong. The Mentat who identifies the weak point in an enemy's system is doing Sun Tzu's work at the computational level.

§22.6 **Cross-reference: Machiavelli on cascades.** Machiavelli's analysis of the fall of states is cascade analysis: a small failure (a bad succession, a neglected fortification) cascades into a larger failure (civil war, foreign invasion). The Mentat who reads Machiavelli is reading cascade analysis in historical form.

CHAPTER 23 — SECOND EFFECTS

§23.1 **The doctrine.** Every action produces a first effect (the one you wanted), a second effect (the one you failed to consider), and a tail of third effects too faint to name. The Mentat who reports only the first effect has committed the most common professional failure of the trade — more common than arithmetic error by an order of magnitude.

§23.2 **The discipline of naming the second effect.** After every recommendation, silence, then: *"The second effect is…"* If nothing comes, the recommendation is incomplete. Common second-effect families to rehearse against until they surface automatically:

- **Relief produces its opposite.** A harsh tax collected now prevents a rebellion later — or breeds one. A clemency shown now prevents martyrdom — or manufactures a legend.
- **Success reorders allies.** Victory creates rivals among your own winners; defeat creates unity among your losers. Count your own side's incentives after any outcome.
- **Warning becomes permission.** Telling a lord of an assassination plot may prevent it, or *normalize it* in his mind until he orders it himself.
- **Measurement distorts.** Reward archers by kills-per-arrow and the archives fill with dead archers' claims and live merchants' forged counts. Whatever you measure will be gamed by someone who understands you better than you understand yourself.
- **Dependence atrophy.** Any capability outsourced to a specialist (a Mentat included) atrophies in the house that buys it. The Mentat who makes his lord dependent on him — wittingly or not — is already halfway to the poison of Chapter 0.5.

§23.3 **The Second-Effect Drill.** During council, in the margin beside each proposal, write one phrase: the second effect. Speak it only when invited or when catastrophic. The private writing trains the muscle; the selective speech preserves your life, because lords of fragile ego hear second effects as insults. (This is not servility; it is the *politics of counsel*, and the student who cannot bear it should take a copper ring and a trading barge instead of the red diamond.)

§23.4 **Cross-reference: Sun Tzu on second effects.** Sun Tzu's warning that "there is no instance of a country having benefited from prolonged warfare" is a second-effect statement: the first effect of war is victory; the second effect is exhaustion. The Mentat who computes the second effect of a military campaign is doing Sun Tzu's work. Sun Tzu's emphasis on speed ("rapidity is the essence of war") is a second-effect consideration: the longer the war, the more second effects accumulate.

§23.5 **Cross-reference: Machiavelli on second effects.** Machiavelli's entire Prince is a second-effect analysis: every recommendation (be feared not loved, be cruel when necessary, keep the people armed) is justified by its second effect (stability, deterrence, loyalty). The Mentat who reads Machiavelli is reading a second-effect manual for princes, and can apply the same analysis to his own counsel.

CHAPTER 24 — THE UNSEEN VARIABLE

§24.1 **Every model is blind somewhere.** The most dangerous element of any forecast is not the error bars; it is the factor *outside the frame entirely* — the unseen variable. History's great houses were not destroyed by their computed enemies but by the factor no one had a column for.

§24.2 **The Siona lesson.** The schools preserve the parable of the daughter who was taught to walk outside the Net of If entirely — unseeable, unmodelled, unkillable by prescience. The parable's computational moral is not mysticism; it is *model humility*: there exist actors, and forces, who by nature or craft fall outside your factor list. Therefore:

- **Red-team on schedule.** Once per major decision, assign a second student (or your second self, at three in the morning) the sole task: *name the factor that, if true, makes this entire computation foolish.* Hunt for it. If found, the computation is rebuilt; if not found, the computation stands with a permanent footnote.
- **Track surprises.** Every quarter, list what actually happened that you did not predict. Not to self-flagellate — to *classify the misses by type*. After four quarters, the pattern of your misses becomes visible: you systematically miss logistics, or personalities, or weather, or the ambitions of second-born sons. Your personal blind-spot map is the single most valuable document you will ever own, and you must never let your patron see it.
- **Respect the exogenous.** Earthquakes, plagues, the whims of a distant emperor, the private grief of an ally: these enter every model as an unmeasured term. The Mentat does not pretend to model them; he *brackets* them — "all figures hold unless Caladan's harvest fails or the Emperor changes his mind, neither of which I can weigh" — and so inoculates his counsel against the appearance of omniscience.

§24.3 **Counter-measure: the checklist of the unseen.** Before sealing any major computation, sweep a short checklist. Each item is a *family* of unseen variables that has destroyed houses before:

1. Logistics (does the plan assume food, water, fuel, and time exist?)
2. Second-order incentives (who gains from the plan failing, among your own?)
3. Communications delay (will the order arrive in the world it was written for?)
4. The single point of failure (one courier, one ford, one advisor, one mood)
5. The enemy's own Mentat (they are also computing; whose model is fresher?)
6. Timing sensitivity (how does the answer degrade per day of delay?)
7. The exhausted variable (is the resource you assume renewable actually finite and near empty?)

Sweeping seven families takes ninety seconds and has a documented habit of changing conclusions.

§24.4 **Cross-reference: Sun Tzu on the unseen.** Sun Tzu's emphasis on "the shape of the enemy" (his dispositions, his habits, his weaknesses) is an unseen-variable search: the commander who has mapped the enemy's unseen variables (his supply lines, his commander's health, his allies' reliability) has eliminated the most dangerous unknowns. Sun Tzu's insistence on "knowing the enemy" is the military version of the Mentat's blind-spot map.

§24.5 **Cross-reference: Machiavelli on the unseen.** Machiavelli's warning that "men are so simple and so ready to obey present necessities that one who deceives will always find those who allow themselves to be deceived" is an unseen-variable statement: the prince who does not account for human gullibility (an unseen variable in his model of his subjects) will be deceived. The Mentat who does not account for his own lord's gullibility will counsel into a void.

CHAPTER 25 — THE MIRROR OF COMPARISON

§25.1 **The comparative method.** When the Net of If is too tangled, the Mentat falls back on the oldest prediction tool: *find the case that already happened and hold it beside this one*. Prediction by mirror is weaker than prediction by mechanism — until the mechanism is beyond your grasp, at which point the mirror is the only honest tool left.

§25.2 **Construction of a mirror.** Three requirements, and they are strict:

1. **Structural similarity, not surface.** Two sieges that share logistics, communications, and relative supply position are mirrors even if one is desert and one is sea. Two sieges that merely both involve castles are not.
2. **Named disanalogies.** Every mirror must list, before use, the ways it *differs* — and each disanalogy must be rated *weighting* (shifts the outcome somewhat) or *fatal* (breaks the comparison entirely). A mirror with no disanalogy list is a drunkard leaning on a painting.
3. **Multiple mirrors.** Three comparisons, minimum, ideally disagreeing. Consensus among mirrors suggests robustness; disagreement among mirrors is information about the decision's true sensitivity.

§25.3 **Differential history as training.** The Schools' favorite exercise: take two historical cases that shared a starting position and diverged at one decision — then ask the student to identify, from the documents available *at the moment of divergence*, which side's advisors should have seen it coming, and from what. This exercise, run weekly for years, produces what no formula produces: **pattern recognition that is honest about its own hindsight.** The student who always finds the answer obvious in retrospect must be trained to notice what *was* knowable then — because a Mentat's entire livelihood is knowing it now, before the retrospect.

§25.4 **The limit of the mirror.** Mirrors fail exactly where the Unseen Variable lives: when the world has moved on to a new kind of case. The Mentat who reaches for one more historical comparison when all mirrors have gone dry is self-soothing. The correct report to the lord is the oldest and hardest sentence in this book:

> *"I have no mirror. I have no factor. I have only the base rate and my own uncertainty, and I tell you plainly: the weight cannot be assigned."*

Masters have died saying this sentence and been remembered for it. Students who have never said it in training will not say it when it matters.

§25.5 **Cross-reference: Sun Tzu on the mirror.** Sun Tzu's historical examples (the battles of antiquity) are mirrors: he compares the present situation to past battles and draws lessons. The Mentat's mirror is the same method, made explicit and disciplined. Sun Tzu's value is not in his conclusions but in his *method* of comparison — the Mentat who reads Sun Tzu is reading a mirror-method manual.

§25.6 **Cross-reference: Machiavelli on the mirror.** Machiavelli's Prince is built on mirrors: every chapter compares the present situation to historical examples (Cesare Borgia, the Roman emperors, the French kings). Machiavelli's method is the Mentat's method: find the historical case, name the disanalogies, draw the lesson. The Mentat who reads Machiavelli is reading a mirror-method manual for princes, and can apply the same method to his own counsel.

CHAPTER 26 — THE MENTAT'S PRESCIENCE

§26.1 **Prescience vs. extrapolation.** The Mentat's extrapolation is probability: "the attack's weight concentrates in the night hours before dawn, three to one." Prescience is *vision*: the ability to see the future directly, not through computation. Paul Atreides, exposed to spice and Bene Gesserit training, achieved prescience — the ability to see all possible futures simultaneously. The Mentat does not have prescience; he has *extrapolation*, which is the human, computational version of the same thing.

§26.2 **The Mentat's prescience-like ability.** Highly skilled Mentats (like the first Duncan Idaho ghola, or Miles Teg), when provided with sufficient data, can deduce probable future outcomes "akin to prescience (though with somewhat less clarity)." This is not mysticism; it is *supralogical hypothesizing* — the naïve mind operating at such depth that the pattern it extracts is indistinguishable from vision. The Mentat who has enough data and enough training can "see" the future in the same way that a chess master "sees" the game: not by computing every move, but by *recognizing the position*.

§26.3 **The danger of prescience-like ability.** The Mentat who begins to believe he has prescience (rather than extrapolation) has fallen into the oracle-shadow (§0.1.2). Prescience is not a Mentat skill; it is a *spice-induced, Bene Gesserit-adjacent* phenomenon that the Mentat must understand but not claim. The Mentat who claims prescience is either addicted to spice (and his computation is compromised) or lying (and his counsel is worthless).

§26.4 **Cross-reference: Sun Tzu on foreknowledge.** Sun Tzu's "foreknowledge cannot be elicited from spirits, nor from gods, nor by analogy with past events, nor from calculations. It must be obtained from men who know the enemy's situation" is a statement about the *source* of prediction: not computation alone, but computation *informed by intelligence*. The Mentat's extrapolation is the computation; the spy's report is the intelligence. Together, they produce something like prescience.

§26.5 **Cross-reference: Machiavelli on fortuna.** Machiavelli's fortuna is the random element that disrupts even the best-laid plans. The Mentat's extrapolation cannot predict fortuna; it can only *bracket* it. Machiavelli's advice to "be impetuous" (to seize fortuna when it appears) is the strategic complement to the Mentat's computation: the Mentat measures the odds; the prince decides whether to bet on fortuna.

CHAPTER 27 — TIMING AND OPPORTUNITY

§27.1 **The temporal dimension.** Every computation has a temporal dimension: the answer that is right today may be wrong tomorrow. The Mentat must compute not just *what* but *when* — when to act, when to wait, when to strike, when to retreat.

§27.2 **The three temporal factors.**

1. **Decay:** information decays. A troop count from last week is less accurate than one from today. The Mentat must weight data by its age.
2. **Urgency:** decisions have deadlines. A decision that must be made in an hour cannot be computed to Rung Three; it must be computed to Rung One and acted on.
3. **Window:** opportunities have windows. The window for action is often shorter than the time needed for full computation. The Mentat must recognize when the window is closing and compute *fast enough*.

§27.3 **The Mentat's timing judgment.** The Mentat does not decide timing; he *reports* it. "The window for action closes in seventy-two hours; after that, the cost of delay exceeds the benefit of further computation." This is the Mentat's contribution to timing: quantifying the trade-off between computation and action.

§27.4 **Cross-reference: Sun Tzu on timing.** Sun Tzu's entire Art of War is a timing manual: "Rapidity is the essence of war," "take advantage of the enemy's unreadiness," "make your way by unexpected routes." The Mentat's timing judgment is the quantitative version of Sun Tzu's timing wisdom. Sun Tzu provides the principle; the Mentat provides the number.

§27.5 **Cross-reference: Machiavelli on timing.** Machiavelli's observation that "the prince who bases his power entirely on words, finding himself without other preparations, is ruined" is a timing statement: the prince who has not prepared *in advance* (when there was time) cannot act *in the moment* (when there is none). The Mentat's daily drill is the preparation; the counsel is the action.

CHAPTER 28 — THE CALCULUS OF POWER

§28.1 **Power as quantity.** Power is not an abstraction; it is a quantity that can be computed. The Mentat who can quantify power can advise his lord with precision; the Mentat who treats power as a mystery is useless.

§28.2 **The power equation.** Power = (Resources × Will × Organization) / (Opposition × Distance × Time). Each factor is computable:

- **Resources:** troops, wealth, territory, technology, allies.
- **Will:** the lord's determination, the people's morale, the army's spirit.
- **Organization:** the efficiency with which resources are deployed.
- **Opposition:** the enemy's power (same formula).
- **Distance:** the distance between the power and its object.
- **Time:** the time available to deploy the power.

§28.3 **The Mentat's power analysis.** The Mentat's power analysis consists of computing each factor, estimating its uncertainty, and reporting the *ratio*. "Our power exceeds theirs by 3:1 in resources, but our will is untested, our organization is poor, and the distance is great. The ratio is favorable but fragile."

§28.4 **Cross-reference: Sun Tzu on power.** Sun Tzu's five fundamentals are a power equation: Moral Law (will), Heaven (timing), Earth (terrain), Commander (organization), Method (discipline). The Mentat's power equation is the quantitative version. Sun Tzu provides the framework; the Mentat provides the numbers.

§28.5 **Cross-reference: Machiavelli on power.** Machiavelli's prince must "be a great estimator" of power — his own and others'. The Mentat's power analysis is this estimation, made precise. Machiavelli's qualitative assessment ("the prince must know how to be a beast") and the Mentat's quantitative assessment ("the prince's power ratio is 3:1, fragile") are the two halves of good counsel.

PART V — STRATEGY AND STATECRAFT

CHAPTER 29 — STRATEGY: THE ART OF WAR CROSS-REFERENCE

§29.1 **Sun Tzu and the Mentat.** Sun Tzu's Art of War is the oldest and most influential manual of strategy. The Mentat who reads it is reading a *companion text* — not a replacement for Mentat training, but a *mirror* that reflects the same principles in strategic language. This chapter maps Sun Tzu's principles to Mentat doctrine.

§29.2 **The five fundamentals as Mentat framework.**

Sun Tzu	Mentat equivalent
Moral Law (harmony between ruler and ruled)	The lord's legitimacy and the people's will — a variable the Mentat must estimate, not assume
Heaven (timing, weather, season)	The temporal factors of §27 — decay, urgency, window
Earth (terrain, distance, danger)	The environmental variables of §34 — terrain, position, logistics
Commander (wisdom, sincerity, benevolence, courage, strictness)	The lord's character — the Mentat's hardest reading (§19.4)
Method and Discipline (organization, supply, logistics)	The power equation's organization factor (§28.2)

§29.3 **Key Sun Tzu principles and their Mentat translations.**

1. **"All warfare is based on deception."** → The frame is the first deception. The Mentat's job is to see through frames, including his own (§5). Deception in data (forged documents, planted rumors) is detected by the Saboteur's Checklist (§40.2).
2. **"If you know the enemy and know yourself, you need not fear the result of a hundred battles."** → The House of Memory (§6) is how the Mentat *knows*: stored, verifiable, retrievable fact. Self-knowledge is the Blind-Spot Map (§24.2). Together, they are the Mentat's version of Sun Tzu's maxim.
3. **"The supreme art of war is to subdue the enemy without fighting."** → The Mentat's highest counsel: find the non-combat solution. The expected-value computation (§17.3) often shows that the cost of fighting exceeds the cost of negotiation. The Mentat who can quantify this is doing Sun Tzu's work.
4. **"Attack is the form of defense."** → In computation: the best defense against a wrong conclusion is an aggressive search for disconfirming evidence. The Mentat who attacks his own conclusion (red-teaming, §24.2) is defending against error.
5. **"In war, the way is to avoid what is strong and to strike at what is weak."** → In computation: avoid the strong (well-supported) conclusions and test the weak (poorly-supported) ones. The Mentat who spends his time verifying what he already believes is wasting his time; the Mentat who tests his weakest assumptions is doing the work.

6. **"He will win who knows when to fight and when not to fight."** → The Mentat's timing judgment (§27). The Mentat who can quantify the window of opportunity is doing Sun Tzu's work.
7. **"He will win who knows how to handle both superior and inferior forces."** → The Mentat's sensitivity pass (§17.4). The Mentat who knows how his computation behaves under varying inputs (superior and inferior data) is doing Sun Tzu's work.
8. **"He will win whose army is animated by the same spirit throughout all its ranks."** → The Mentat's counsel must be *aligned* with the lord's organization. A computation that the organization cannot execute is worthless. The Mentat must compute not just the optimal strategy but the *executable* strategy.
9. **"He will win who, prepared himself, waits to take the enemy unprepared."** → The Mentat's daily drill (§38). Preparation is the Mentat's core discipline; the unprepared Mentat is the defeated Mentat.
10. **"He will win who has military capacity and is not interfered with by the sovereign."** → The lord must not override the Mentat's computation. A lord who interferes has not gained a computation; he has lost a Mentat. This is the Mentat's most important political principle, and the hardest to enforce.

§29.4 **Sun Tzu's nine situations.** Sun Tzu's nine situations (dispersive, facile, contentious, open, ground of intersecting highways, serious, difficult, hemmed-in, desperate) are a *classification system* for strategic positions. The Mentat can use this classification to *frame* a military problem: "We are in a 'serious' situation — deep in enemy territory, with no retreat. The computation changes accordingly."

§29.5 **Sun Tzu's spies.** Sun Tzu's five types of spies (local, inward, converted, doomed, surviving) are an *intelligence taxonomy*. The Mentat who understands this taxonomy can organize his information-gathering and compute the reliability of each source. The converted spy (an enemy spy turned) is the most valuable and the most dangerous — the Mentat's independence check (§16.3) is essential when using converted-spy intelligence.

§29.6 **Cross-reference: Machiavelli and Sun Tzu.** Machiavelli's prince and Sun Tzu's general share a common insight: *war is politics by other means, and politics is war by other means*. The Mentat who serves both a prince and a general must translate between the two frames: the prince's frame is power and survival; the general's frame is victory and logistics. The Mentat's computation must serve both, and the two frames sometimes conflict.

CHAPTER 30 — STATECRAFT: MACHIAVELLI CROSS-REFERENCE

§30.1 **Machiavelli and the Mentat.** Machiavelli's The Prince is the oldest and most influential manual of statecraft. The Mentat who reads it is reading a *companion text* — not a replacement for Mentat training, but a *mirror* that reflects the same principles in political language. This chapter maps Machiavelli's principles to Mentat doctrine.

§30.2 **Key Machiavelli principles and their Mentat translations.**

1. **"It is better to be feared than loved, if you cannot be both."** → The Mentat's counsel on the lord's *image*: the lord's power depends on perception, and the Mentat must compute the trade-off between fear and love in his lord's specific situation. This is not the Mentat's decision; it is data for the lord's decision.
2. **"The ends justify the means."** → The Mentat does not endorse this; he *recognizes* it. When a lord acts on this principle, the Mentat must compute the *second effects* (§23) of the means, because the means always produce consequences that the ends do not account for.
3. **"Fortune favors the bold."** → Machiavelli's fortuna is probability. The bold action is the one with high variance: high potential gain, high potential loss. The Mentat's job is to quantify the variance, not to forbid the gamble.
4. **"A prince must be a fox and a lion."** → The Mentat is the fox (cunning, perception) serving the lion (force, will). The Mentat's fox-nature (§1.9) is his professional identity; he must not try to be the lion, because the lion's job is the lord's.
5. **"Appearances vs. reality."** → The naïve mind (§3) sees through appearances. The Mentat who reads Machiavelli is reading a manual on the gap between appearance and reality, and the Mentat's skill is to *quantify* that gap.
6. **"The new prince and the hereditary prince."** → The new prince faces higher risk and must act more decisively; the hereditary prince can afford caution. The Mentat's computation must account for the prince's *type*: a new prince needs bold counsel; a hereditary prince needs conservative counsel.
7. **"Military organization: citizen militia, not mercenaries."** → The Mentat's power analysis (§28) must account for troop *quality*: citizen militias fight harder (higher will) but are less skilled; mercenaries are skilled but unreliable. The Mentat who computes troop strength without accounting for troop type is computing a phantom.
8. **"Cruelty well-used vs. cruelty ill-used."** → The Mentat's second-effect analysis (§23) of cruelty: well-used cruelty is swift, decisive, and followed by benefits; ill-used cruelty is prolonged, arbitrary, and generates hatred. The Mentat quantifies the *duration* and *targeting* of cruelty.
9. **"Liberality and parsimony."** → The Mentat's resource analysis: a lord's generosity is a resource expenditure, and the Mentat must compute its return. Machiavelli's point that liberality leads to poverty (which leads to hatred) is a second-effect computation.

10. **"The prince and his ministers."** → The Mentat is a minister. Machiavelli's advice on ministers (choose wise ones, not flattering ones; reward them, but not too much; keep them dependent) is advice about the Mentat's *position*. The Mentat must understand his position to serve well.
11. **"Flatterers."** → The Mentat's greatest enemy is not the enemy outside the house; it is the flatterer inside it. The Mentat must compute the *influence* of flatterers on the lord's decisions and report it. This is the most dangerous counsel a Mentat can give, because it implies the lord is being manipulated — and the lord may prefer the manipulation to the truth.

CHAPTER 31 — DECEPTION AND DISGUISE

§31.1 **Deception as computation.** Deception is the art of making another's model of the world diverge from reality. The Mentat who understands deception can both *detect* it (in others) and *compute its effects* (when his lord employs it). The Mentat does not practice deception himself — his vow forbids it — but he must understand it as thoroughly as a physician understands disease.

§31.2 **The four deceptions.**

1. **Concealment:** hiding a fact. The Mentat detects it by noting what is *absent* from a dossier (§40.2).
2. **Distortion:** presenting a fact in a misleading way. The Mentat detects it by checking the *source* and *context* of each fact.
3. **Fabrication:** inventing a fact. The Mentat detects it by cross-referencing and by noting *precision* (forgeries are often too precise).
4. **Misdirection:** drawing attention to one fact to hide another. The Mentat detects it by asking "what is the *purpose* of this information being given to me?"

§31.3 **Cross-reference: Sun Tzu on deception.** Sun Tzu's entire Art of War is a deception manual: "All warfare is based on deception. Hence, when able to attack, we must seem unable; when using our forces, we must seem inactive; when we are near, we must make the enemy believe we are far away." The Mentat who reads Sun Tzu is reading a deception manual, and can apply the same analysis to data (what is the data *trying* to make me believe?) and to people (what is this person *trying* to make me think?).

§31.4 **Cross-reference: Machiavelli on deception.** Machiavelli's prince must "be a great pretender and dissembler" — deception is a tool of statecraft. The Mentat who serves a Machiavellian lord must compute the *second effects* of the lord's deceptions, because deceptions that succeed produce *trust erosion* that eventually undermines the lord's power. The Mentat who can quantify this erosion is doing the most valuable (and most dangerous) counsel.

CHAPTER 32 — INTELLIGENCE AND SPIES

§32.1 **The Mentat as intelligence officer.** In the Imperium, intelligence is the lifeblood of statecraft. The Mentat who can organize, evaluate, and compute from intelligence is worth more than a regiment. The Mentat is not a spy — he is the *processor* of spy-gathered data.

§32.2 **Sun Tzu's five spies.** Sun Tzu's taxonomy:

1. **Local spies:** inhabitants of the enemy's territory.
2. **Inward spies:** enemy officials.
3. **Converted spies:** enemy spies turned.
4. **Doomed spies:** spies given false information to pass to the enemy.
5. **Surviving spies:** spies who return with information.

The Mentat's job is to *evaluate* each spy's report: what is the spy's type? What are his incentives? What is his reliability? The Mentat who treats all spy reports as equal has not understood intelligence.

§32.3 **The reliability computation.** For each intelligence report, compute:

- **Source reliability:** how trustworthy is the source? (Based on past accuracy, access, incentives.)
- **Information credibility:** how plausible is the information itself? (Based on consistency with other data, physical possibility, base rates.)
- **Corroboration:** is the information supported by independent sources?

Report: "Source: inward spy, reliability 60%. Information: plausible, consistent with other data. Corroboration: none. Confidence: low. Recommendation: seek corroboration before acting."

§32.4 **Cross-reference: Machiavelli on intelligence.** Machiavelli's prince must "have good spies" and "know how to discover the designs of his enemies." The Mentat's intelligence evaluation is this discovery, made systematic. Machiavelli's emphasis on *personal* knowledge of the enemy's intentions (rather than reliance on formal intelligence) aligns with the Mentat's emphasis on person-reading (§19).

CHAPTER 33 — ALLIANCE AND BETRAYAL

§33.1 **The mathematics of alliance.** An alliance is a non-zero-sum game in which two parties cooperate for mutual benefit. The Mentat must compute: what does each party gain? What does each party risk? What is the duration of the alliance? What are the conditions for dissolution?

§33.2 **The betrayal computation.** Betrayal occurs when one party's expected gain from defection exceeds the expected gain from cooperation. The Mentat computes this for *both* sides: "Our ally's expected gain from betraying us is X; his expected gain from staying is Y. If X > Y, betrayal is rational. We must either increase Y (make staying more valuable) or decrease X (make betraying more costly) or both."

§33.3 **Cross-reference: Sun Tzu on alliances.** Sun Tzu: "In all fighting, the direct method may be used for joining battle, but indirect methods will be needed in order to secure victory." Alliances are indirect methods: they secure victory without fighting. The Mentat who can compute the value of an alliance is doing Sun Tzu's work.

§33.4 **Cross-reference: Machiavelli on alliances.** Machiavelli: "A prince who is not wise himself cannot be well advised" — and a prince who allies without computing the ally's incentives has not been well advised. Machiavelli's emphasis on the *instability* of alliances (they last only as long as mutual interest lasts) is the Mentat's betrayal computation in qualitative form. The Mentat quantifies what Machiavelli observes.

CHAPTER 34 — TERRAIN, ENVIRONMENT, AND POSITION

§34.1 **Terrain as variable.** Terrain is not scenery; it is a *computational variable* that affects every other factor. The Mentat who ignores terrain is computing on a flat map; the Mentat who incorporates terrain is computing on the real ground.

§34.2 **Sun Tzu's terrain types.** Sun Tzu classifies terrain into six types:

1. **Accessible:** both sides can traverse freely. Occupy the high ground first.
2. **Entangling:** easy to enter, hard to exit. Do not be lured in.
3. **Temporizing:** neither side gains advantage. Do not fight; negotiate.
4. **Narrow passes:** few can enter; control them, and the enemy cannot pass.
5. **Precipitous heights:** control them, and the enemy below is at your mercy.
6. **Great distance:** far from both sides; the first to arrive wins.

Each type changes the computation: force ratios, supply lines, communication delays, and morale all vary with terrain.

§34.3 **The Mentat's terrain analysis.** For any military problem, the Mentat computes:

- **Force ratio adjustment:** how does terrain affect the effective force ratio? (A narrow pass neutralizes numerical superiority.)
- **Supply line length:** how long are the supply lines? (Long supply lines decay; short ones sustain.)
- **Communication delay:** how long does an order take to reach the front? (Terrain affects speed.)
- **Retreat options:** if the battle goes wrong, what are the retreat routes? (No retreat = desperate situation.)

§34.4 **Cross-reference: Machiavelli on terrain.** Machiavelli's emphasis on fortresses and terrain ("a prince must... be a great estimator of terrain") aligns with the Mentat's terrain analysis. Machiavelli's observation that fortresses are "useless" without a loyal population is a terrain insight: terrain (fortresses) is secondary to will (loyalty).

CHAPTER 35 — TIMING AND TEMPO

§35.1 **Tempo as weapon.** Tempo — the speed at which decisions are made and executed — is a strategic weapon. The side that decides faster, even if its decisions are slightly worse, often wins, because the slower side is always reacting, never initiating.

§35.2 **The Mentat's tempo computation.** The Mentat computes the *tempo gap*: how fast can we decide vs. how fast can the enemy decide? If our tempo is faster, we can force the enemy into reactive mode, where his decisions are constrained by ours. If our tempo is slower, we must either improve it or find a way to slow the enemy.

§35.3 **Cross-reference: Sun Tzu on tempo.** Sun Tzu's "rapidity is the essence of war" is a tempo statement. The Mentat's tempo computation is the quantitative version: how rapid is rapid enough? Sun Tzu provides the principle; the Mentat provides the number.

§35.4 **Cross-reference: Machiavelli on tempo.** Machiavelli's emphasis on *decisiveness* ("it is better to be impetuous than cautious") is a tempo statement. The prince who hesitates loses the window; the prince who acts (even imperfectly) seizes it. The Mentat's role is to ensure that the prince's decisiveness is *informed* — that he acts on computed probabilities, not on impulse.

CHAPTER 36 — THE MENTAT IN COUNCIL

§36.1 **The council as arena.** The Mentat's primary workplace is the lord's council. In council, the Mentat must: present his computation, defend it against challenge, detect the biases of other counselors, and read the lord's reaction. This is the Mentat's highest-stakes environment.

§36.2 **The council protocol.**

1. **Present the computation first.** Before any debate, state the figures, their labels, their brackets, and the load-bearing branch. This establishes the factual baseline before opinion enters.
2. **Invite challenge.** "These are my figures. I ask the council to test them." This demonstrates confidence and invites the correction that the Mentat cannot find alone.
3. **Listen for bias.** Each counselor has interests. The Mentat notes which counselors attack the computation and which attack the conclusion — the former are testing, the latter are defending interests.
4. **Read the lord.** The lord's reaction to the computation (not his words, his body) tells the Mentat whether the lord has already decided. If so, the Mentat's computation is now *post-hoc justification*, and he must decide whether to provide it or refuse.
5. **Speak the second effect.** If the lord is about to decide, the Mentat speaks the second effect (§23.2) — once, clearly, then falls silent.

§36.3 **The Mentat's council discipline.**

- Never argue with a counselor who is defending an interest; the argument is not about the computation, it is about the interest.
- Never repeat a computation that has been rejected; recompute and present fresh.
- Never let a council run past the point where the sphere can hold the problem; when the sphere dims, call for a recess.
- Always leave the council with a written summary of what was decided and what was left open.

§36.4 **Cross-reference: Sun Tzu on council.** Sun Tzu's emphasis on "the general who wins a battle makes many calculations in his temple before the battle is fought" applies to council: the Mentat who computes before council, not during it, wins. The council is for *presentation* and *challenge*, not for *computation*.

§36.5 **Cross-reference: Machiavelli on council.** Machiavelli's advice on ministers (choose wise ones, not flattering ones) applies to the council: the Mentat should assess the *quality* of the council's advice, not just its content. A council of flatterers produces bad counsel regardless of the Mentat's computation.

CHAPTER 37 — THE MENTAT IN CRISIS AND WAR

§37.1 **Crisis computation.** In crisis, the Mentat must compute faster, with less data, and under higher stakes. The principles:

- **Drop to Rung One.** In crisis, order-of-magnitude is usually sufficient; precision is a luxury.
- **Shorten the sphere.** Hold only the three most critical factors; let the rest go.
- **Speak in brackets.** "Between X and Y" is more useful in crisis than "X" — it acknowledges uncertainty without paralyzing.
- **Name the decision.** "The decision is: attack or retreat. The computation favors attack 3:1, but the uncertainty is high." This frames the crisis as a decision, not a computation.

§37.2 **War computation.** In war, the Mentat's computation is continuous: troop positions, supply levels, enemy movements, weather, morale. The Mentat must maintain a *live model* — updated as new data arrives, with explicit tracking of what has changed and what hasn't.

§37.3 **The Mentat's war-time discipline.**

- Update the model at least daily; hourly in active combat.
- Track the *delta*: what changed since the last update? What caused the change?
- Flag *surprises*: data that contradicts the model. Surprises are the most valuable data in war.
- Maintain the *Unknowns shelf*: what do we not know? In war, the unknowns are often more important than the knowns.

§37.4 **Cross-reference: Sun Tzu on war.** Sun Tzu's entire Art of War is a war-time computation manual. The Mentat who has internalized Sun Tzu's principles can compute in the language of war: "The enemy's force ratio is 2:1 in our favor, but his terrain advantage is significant, and his supply line is shorter. Net: favorable but not decisive." This is Sun Tzu's assessment translated into Mentat computation.

§37.5 **Cross-reference: Machiavelli on war.** Machiavelli's emphasis on *decisive action* ("it is better to be impetuous than cautious") applies to war-time computation: the Mentat must compute fast enough to support decisive action, but not so fast that the computation is wrong. The balance between speed and accuracy is the Mentat's hardest war-time judgment.

PART VI — THE STUDENT'S LIFE

CHAPTER 38 — THE DAILY REGIMEN

§38.1 **The shape of the day.** The regimen below is the Salusan standard, adjusted by most schools. Hours are relative, not absolute — what matters is the *order* and the *spacing*.

Period	Work
Rising	Three-breath descent (§2.2); Litany; sphere-drill on a single quality, 60 seconds ×3
First block (~90 min)	**Deep computation** — hardest material of the day, before the world claims you. No messages. No eyes on others' business.
Interval	Physical labor or walking. The sphere must empty fully; it refills only when it has emptied.
Second block (~60 min)	**House work** — Returning Drills due that day (§8.2); new loads to loci
Interval	Food, and the second inventory (§1.3) taken honestly
Third block (~60 min)	**Chains and speed drills** — backward recitation, complements, casting-out under mild alarm (§7.5)
Afternoon	**Counsel practice** — write one page of mock counsel to a given lord: three figures, one bracket, one second effect, one line beginning "The second effect is…"
Evening	**The Mirror** — one historical comparison, disanalogies named (§25.2)
Before sleep	Ledger: five calibrated predictions recorded (§16.5). Sphere-drill. Sleep is not optional; a Mentat on short sleep is a confabulation machine with good manners.

§38.2 **The weekly variation.** One day per week: no drills at all — instead, *walk in unfamiliar country*, preferably with people who do not know what you are. The House of Memory grows best from fresh experience; a student who drills without living fills his fort with furniture he has never sat upon.

§38.3 **The body is infrastructure.** The Schools are unromantic about this: blood-sugar, hydration, sleep, and injury are *computational variables*. A Mentat who treats the body as a mule to be whipped will produce, in his thirties, the most dangerous product in the trade — slow, fluent, confident error. Train the body as you train the number.

§38.4 **Cross-reference: Sun Tzu on the regimen.** Sun Tzu's emphasis on "the general who wins a battle makes many calculations in his temple before the battle is fought" applies to the daily regimen: the Mentat who drills daily is making his calculations in the temple; the Mentat who does not is trying to compute in the battle. The regimen is the temple.

§38.5 **Cross-reference: Machiavelli on the regimen.** Machiavelli's prince must "accustom his body to hardships" and "study the art of war." The Mentat's regimen is the same: the body must be trained, the mind must be trained, and both must be trained *together*, because a weak body produces a weak mind.

CHAPTER 39 — THE TRIALS

§39.1 **The three rings.** Examinations occur in three rings, each sealed from the next:

Ring One — Recall. A document is read aloud once (or a House of Memory is interrogated at random loci). The student recites, then answers five analytical questions, then appends to each answer its label: *exact*, *reconstructed*, *estimated*, *unknown*. Failure to label is failure in itself.

Ring Two — Computation. Twenty problems, forty minutes, no stylus — all figures carried in the sphere and written only at the end. Problems span: complements and decomposition (§12), one near-base multiplication, one log-skeleton multiplication, one checkpoint probability (§16.4), one proportional scaling with named exponent (§13), one Fermi estimate with *two independent routes* and a bracket (§14). Partial credit exists only for *method*; a right answer from an unverifiable method is graded as a fail, deliberately — this is the Schools refusing to train luck.

Ring Three — Counsel. A scenario is given in writing: a lord's problem, with a dossier of twelve documents of unequal reliability (two are forgeries; the student is never told how many). The student produces one page: the decision-relevant figures, each labelled; a bracket; a scenario tree to three rungs; the load-bearing branch; the second effect; and one sentence of what is *not known*. Graded on structure and honesty, not on the conclusion.

§39.2 **Self-trial (weekly, unproctored).** Every seventh day, alone: thirty minutes of Ring Two conditions, then grade yourself against the key with the mercilessness of a proctor. Then — the part students skip — write one paragraph on *why* each error occurred: haste, method gap, misread, sphere-collapse, or confabulation. Error causes cluster. Yours are already clustering; find them in the first month rather than the fifth year.

§39.3 **Cross-reference: Sun Tzu on examination.** Sun Tzu's emphasis on "making many calculations" before war is the Mentat's Ring Two: the student who has drilled the computations can produce them under pressure; the student who has not cannot. The Trials are the temple; the field is the battle.

§39.4 **Cross-reference: Machiavelli on examination.** Machiavelli's observation that "a prince must... be a great estimator" applies to the Trials: the Mentat who can estimate under pressure (Ring Two, Rung One) is the Mentat who can serve in crisis. The Trials test not just knowledge but *performance under load*.

CHAPTER 40 — ERRORS, SABOTAGE, AND THE CHECKLIST

§40.1 **The taxonomy of Mentat errors.** Memorize the seven:

1. **Arithmetic slip** — caught by casting-out (§12.6); the least shameful.
2. **Method gap** — wrong tool for the quantity; fixed by breadth of drill.
3. **Frame error** — right arithmetic inside a false frame (§5.1); the most common source of catastrophe.
4. **Prior neglect** — multiplying clues without base rates (§16.2); produces dramatic, confident, wrong counsel.
5. **Exponent blindness** — linear scaling of square-law phenomena (§13.2).
6. **Sphere collapse** — continuing work after the sphere has dimmed; produces phantom factors (§2.4).
7. **Confabulation** — reporting as memory what was invented; **the one unforgivable error.** It is forgivable once, as a student, if confessed immediately and retrained from zero. Concealed once, and the student is dismissed — not for the lie, but because the Schools cannot certify a mind that no longer knows which of its outputs are its own.

§40.2 **The Saboteur's Checklist.** An old maxim of the trade: *assume the dossier is fed to you by someone who wants a particular answer.* Before sealing counsel, sweep:

1. What in this dossier is *undated*? (Undated facts are not facts; they are traps with patient fuses.)
2. What arrives *too conveniently* to support the conclusion my patron wants? (Desire farms forgeries.)
3. What is *absent* that any honest report would contain?
4. Which number in the dossier is *precise to a suspicious degree*? (Precision is a costume; forgers wear it.)
5. Who *cannot benefit* from this conclusion? (Their silence is evidence.)
6. If I invert the conclusion, which documents would I expect to see — and are any of them here?
7. Who delivered this dossier, and what did they gain if I say yes?

§40.3 **Self-sabotage.** The checklist exists to defend against others; the harder sweep is against oneself. Before counsel is spoken, ask: *what do I want, and where in this computation has my wanting left fingerprints?* Then find the fingerprints by force. A Mentat cannot purge wanting — but he can *account* for it, and an accounted bias is half neutralized.

§40.4 **Cross-reference: Sun Tzu on sabotage.** Sun Tzu's emphasis on "knowing the enemy" includes knowing the enemy's *deceptions*. The Saboteur's Checklist is the Mentat's version of Sun Tzu's enemy-analysis: the Mentat who sweeps the checklist is checking for the enemy's (or the dossier-feeder's) deception.

§40.5 **Cross-reference: Machiavelli on sabotage.** Machiavelli's warning that "men are so simple and so ready to obey present necessities that one who deceives will always find those who allow themselves to be deceived" is the foundation of the Saboteur's Checklist: the Mentat who does not sweep for deception is

allowing himself to be deceived.

CHAPTER 41 — THE ETHICS OF THE MENTAT

§41.1 **Service, not rule.** The Mentat serves. The serving-Mentat who begins to prefer outcomes has begun the slide. The classic form of the slide is subtle: "my lord will choose X; X is worse; therefore I shall shade the figures so the choice falls on Y." The shading is always small. The first shading is always small. The Schools' position is absolute: **shade a figure once and you have become the ruler wearing your lord's clothes, and the record of men who have worn both poorly is long and cold.**

§41.2 **The formula of counsel.** The proper shape of the report, learned verbatim by every student:

> *"These are the figures. These are their labels and their brackets. This is the branch on which the decision turns. This is the second effect I am most able to see. This is what I do not know. The choice is yours."*

The last sentence is not courtesy. It is the load-bearing wall of the entire profession. Say it even when the lord has already decided and merely wants your number to hide behind — *especially* then, because that is when the number will be quoted back at the funeral.

§41.3 **Refusals.** A Mentat may refuse an assignment on three grounds, and only three: the task is beyond his trained region (and he must say so); the task requires him to shade or confabulate; the task serves a purpose he judges monstrous *beyond the scope of ordinary service* — and for the third, the Mentat must accept the consequences of refusal, which in the histories have not been gentle. Note carefully: a Mentat does not refuse to serve a *flawed* lord; flawed lords are the only kind history issues. The refusal is about the *work*, not the man.

§41.4 **The two exemplars — a study in contrast.** The Schools teach every student the pair:

- **The loyal instrument.** Trained, exact, bracketed, honest about uncertainty — and yet: obedience carried past the point of judgment becomes the right hand of catastrophe. Loyalty is not absolution. The Mentat who obeyed every order and never once said *"my lord, this is unwise, and here is why"* shares the weight of what followed.
- **The corrupted instrument.** Trained, exact, and *ambitious* — computing not for the lord but for the throne beside him; shaping figures to accelerate the lord's fall; the poisoner, the flatterer, the one who sold counsel by the drop. The corruption does not begin with malice. It begins with *pleasure in being the smartest man in the room* — a pleasure every student of this book will feel, and must therefore watch for, for the rest of his career.

Neither exemplar is a cautionary tale about arithmetic. Both are cautionary tales about *the relationship between knowledge and power*, and the answer the Schools give is the same in both cases: the Mentat's power is real, therefore the Mentat's discipline must be absolute, therefore *the last sentence of counsel is never omitted.*

§41.5 **Cross-reference: Sun Tzu on ethics.** Sun Tzu's emphasis on "the general who wins a battle makes many calculations in his temple before the battle is fought" includes the calculation of *whether the battle should be fought at all*. The Mentat's ethics is this calculation: is this computation serving the living, or is it

serving the Mentat's ambition? Sun Tzu's warning that "no country has ever benefited from prolonged warfare" is an ethical statement: the general who prolongs war for personal glory is not a general but a butcher.

§41.6 **Cross-reference: Machiavelli on ethics.** Machiavelli's prince operates in a world where "the ends justify the means," but the Mentat does not. The Mentat's ethics is the *limit* on Machiavellian statecraft: the prince may choose the means, but the Mentat must compute their *second effects* (§23) and report them. The Mentat who fails to report the second effects of the prince's means has become the prince's accomplice, not his advisor.

CHAPTER 42 — THE MENTAT'S END

§42.1 **Aging and the Mentat.** The Mentat's abilities decline with age — not the arithmetic (which is overlearned and persists), but the *sphere* (which requires energy and declines) and the *naïve mind* (which requires openness and tends to rigidify). The aging Mentat must recognize his decline and adjust: shorter sessions, more reliance on the House, more delegation to younger Mentats.

§42.2 **Retirement.** The Schools encourage retirement at the point where the Mentat can no longer hold the sphere for a full session. Retirement is not disgrace; it is *honesty*. The Mentat who retires and trains successors has served the Order; the Mentat who clings to his position past his capacity has not.

§42.3 **Succession.** The Mentat's most important legacy is his successor. The Schools encourage Mentats to identify and train their successors years before retirement, so that the transition is gradual and the House's knowledge is transferred. The Mentat who dies without a successor has left a gap that the Order must fill from outside — and outside Mentats are never as good as home-grown ones.

§42.4 **The red diamond.** When the student is finally marked — the small red sigil at the temple that tells the galaxy what walks inside the skull — the Master speaks only one line:

> *"Compute for the living. Doubt yourself for them. Serve them the truth, whole and unshaded, or return the ring."*

Then the student speaks the maxim of Chapter 0.1, §0.1.6, one last time — aloud, before witnesses, as architecture rather than speech:

> *The Mentat computes. The Mentat weights. The Mentat speaks. The Mentat does not choose.*

Go and be useful.

APPENDIX A — TABLES FOR THE MIND

A.1 Primes below 200 (chunk into 100s): 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181 191 193 197 199

A.2 Squares 1–25 (the rest compose: $(a \pm b)^2$): 1 4 9 16 25 36 49 64 81 100 121 144 169 196 225 256 289 324 361 400 441 484 529 576 625

A.3 Powers of 2 to 2^{16}: 1 2 4 8 16 32 64 128 256 512 1024 2048 4096 8192 16384 32768 65536

A.4 Common constants: $\sqrt{2} \approx 1.414$ · $\sqrt{3} \approx 1.732$ · $\pi \approx 3.142$ · $e \approx 2.718$ · $g \approx 9.8$ · $1 \ln 10 \approx 2.303$ · 1 yr ≈ $\pi \times 10^7$ s

A.5 Log skeleton — see §15.2; do not skip the verbatim commitment.

A.6 Reciprocals to 1/20 (as decimals): 1.0 0.5 0.333 0.25 0.2 0.167 0.143 0.125 0.111 0.1 0.091 0.083 0.077 0.071 0.067 0.063 0.059 0.056 0.053 0.05

A.7 Sun Tzu's five fundamentals (for quick reference): Moral Law · Heaven · Earth · Commander · Method and Discipline

A.8 Machiavelli's prince's virtues (for quick reference): Lion (force) · Fox (cunning) · Be feared not loved · Be impetuous not cautious · Keep the people armed · Choose wise ministers not flatterers

A.9 The Mentat's five standard reframes (for quick reference): Advocate · Loser · Enemy · Child · Bookkeeper

A.10 The Saboteur's Checklist (for quick reference): Undated · Too convenient · Absent · Suspiciously precise · Who cannot benefit · Invert the conclusion · Who delivered

APPENDIX B — THE EXERCISE BANK

Grade I (weeks 1–4). Complements at bases 10/100/1000, twenty per day. Digit-sums and casting-out-9s checks, twenty per day. Three-breath descent to three consecutive cleans. House of Memory: build and map fifty loci. Naïve observation drill (§3.4) weekly.

Grade II (weeks 5–12). Decomposition multiplication, thirty per day, under five minutes total. Peg-lists of ten items, forward and backward, five per week. Returning Drill schedule begun on twenty loaded facts. Ten calibrated predictions per week. Frame-multiplication drill (§1.6) daily, ten frames. Reading protocol (§11.5) on one document daily.

Grade III (months 4–9). Near-base multiplication across three bases. Log-skeleton arithmetic: five products per week by logs, ±2% required. One Fermi estimate per week, two routes, bracket, factor-list. Chains of eight links, forward and backward, under brisk walking. Digit-string chunking: strings of twelve, then sixteen. Mirror exercise (§25.3) monthly.

Grade IV (months 10–24). Checkpoint-probability problems from live (or archived) intelligence. Scenario trees to three rungs with sensitivity passes. Mirrors: three historical comparisons monthly, disanalogy lists graded. Red-team sessions against peers' completed computations. Full mock Rings One through Three, monthly. Counsel practice (§38.1) daily. Saboteur's Checklist (§40.2) on every dossier.

Cross-reference exercises. Monthly: take one Sun Tzu principle and one Machiavelli principle, and write a one-page Mentat analysis showing how both apply to a current problem. This trains the cross-reference instinct that makes Part V useful rather than ornamental.

APPENDIX C — GLOSSARY

The Sphere — the bright interior space in which live factors are held during deep work. **The Stance** — the physical and mental posture of neutral readiness. **The House of Memory** — the student's constructed loci-system. **Scrubbing** — the seasonal dismantling of dead loci and stale facts. **The Chain** — an association-sequence linking events or quantities by vivid causal images. **The Returning Drill** — the spaced-review schedule of §8.2. **The Ladder** — three-rung scheme of estimation: order of magnitude, two figures, full precision. **The Broken Sphere** — double independent estimation by differing factorizations. **The Skeleton** — the verbatim log-table held in the House. **The Net of If** — the conditional structure of forecast futures; its heavy cords are base rates. **Second Effect** — the first consequence of an action that the actor failed to consider. **The Unseen Variable** — the factor outside the model's frame entirely. **The Mirror** — a historical comparison, with named disanalogies, used for prediction. **The Saboteur's Checklist** — the seven-point dossier-interrogation of §40.2. **The Label** — the mandatory tag appended to each output: exact, reconstructed, estimated, unknown. **Confabulation** — memory reporting invention as fact; the unforgivable error. **The Red Diamond** — the Mentat's mark. **The Naïve Mind** — the supralogical faculty of perception without preconception. **The Mentat's Curse** — the inability to stop computing; the neurological cost of training. **Sapho** — the addictive stimulant that doubles or triples Mentat processing power; stains the lips. **Melange** — the spice; extends life, heightens awareness, produces prescience in large doses. **Twisted Mentat** — a Mentat conditioned without ethical restraints; the Tleilaxu perversion. **Prescience** — direct vision of the future; beyond Mentat extrapolation; spice/Bene Gesserit-linked. **Kwisatz Haderach** — the "shortening of the way"; a being who sees both ancestral memories; beyond Mentat. **Fortuna** — Machiavelli's personification of probability; the random element in plans. **Virtù** — Machiavelli's term for the prince's skill, courage, and adaptability. **Shi** — Sun Tzu's concept of strategic advantage; the configuration of forces. **Ji** — Sun Tzu's "calculations"; the assessment of factors before war. **The Fox** — Machiavelli's cunning advisor; the Mentat's professional identity. **The Lion** — Machiavelli's forceful prince; the lord's role, not the Mentat's.

APPENDIX D — ANSWER KEY

§12.2 Complements: 63; 352. (Drill until reflex; check by re-adding.) **§12.3 Labors:** 924; 236; 2268; 27. **§12.4:** 9118; 990021. **§12.5:** 2304; 1225; 4225. **§12.6 Casting-out demo:** 47→2, 83→2, product residue 4; 3901→4 ✓. **§13.4 Brackets:** any defensible ×½/×2 (or tighter with cause) accepted; no-bracket = fail. **§14.3 Spice estimate:** order 10^5, bracket 5×10^4–1.2×10^5, levers named. (Route disagreement >2× triggers third route.) **§15.4 Log walk:** ≈3350 (true 3358). **§15.6:** ≈6.9×. **§16.4 Checkpoint:** 31% (90 of 288 accusations true). **§17.5 Tree — instructor's solution sketch:**

- Branches: enemy size (3 values × ⅓), ford passability (2 values, 0.5/0.5 at mid-estimate), irregulars (0.6/0.4).
- Best case: enemy 2500, ford passable, irregulars join → favorable engagement, expected campaign value **positive**.
- Worst case: enemy 3500, ford delayed (re-crossing under fire), irregulars refuse → **severe loss**.
- Expected value sits near break-even and is *negative if irregular support is removed* — the load-bearing branch is **ally fidelity** (sensitivity: ±25% swing in irregular odds flips the sign of expected value). The counsel therefore pivots on verifying irregular intent, not on further scouting of enemy numbers. **§24.3 Checklist sweep** (graded for completeness, not for one right answer). **§39.2** — self-graded; error-cause paragraph is mandatory.

APPENDIX E — CROSS-REFERENCE INDEX

Dune → Mentat doctrine:

- Butlerian Jihad → §0.2 (commandment as assignment)
- Erasmus/Gilbertus Albans → §0.2.2, §0.4 (founding, grades)
- Thufir Hawat → §0.7.2 (loyal instrument)
- Piter de Vries → §0.5, §0.7.3 (twisted Mentat, corrupted instrument)
- Mentat's Curse → §0.6.1
- Sapho juice → §0.6.2
- Melange/spice → §0.6.3
- Kwisatz Haderach → §0.6.4
- Mentat's Doubt → §0.6.5
- Naïve mind → §0.1.7, §3
- First Law of Mentat → §4.2
- Mentat Mantra → §4.3
- Twisted Mentats → §0.5
- Mentat Schools → §0.3, §0.4
- Leto II's suppression → Part Zero (context)
- Miles Teg → §26.2
- Duncan Idaho ghola → §26.2
- Siona → §24.2

Art of War → Mentat doctrine:

- Five fundamentals → §1.8, §29.2
- Commander qualities → §1.8
- Sovereign must not interfere → §1.8, §29.3.10
- Terrain → §3.5, §34
- Water metaphor → §18.5
- Deception → §5.6, §29.3.1, §31.3
- Know enemy/self → §6.8, §29.3.2
- Win without fighting → §29.3.3
- Attack is defense → §29.3.4
- Avoid strong, strike weak → §22.5, §29.3.5
- Timing → §27.4, §29.3.6
- Superior/inferior forces → §29.3.7

- Same spirit throughout → §29.3.8
- Prepared waits unprepared → §29.3.9
- Calculations in temple → §12.8, §38.4
- Nine situations → §29.4
- Five spies → §29.5, §32.2
- Rapidity → §23.4, §35.3
- Foreknowledge → §26.4
- Spies and memory → §8.5

The Prince → Mentat doctrine:

- Fox and lion → §1.9, §30.2.4
- Feared vs. loved → §30.2.1
- Ends justify means → §30.2.2
- Fortune favors bold → §16.7, §30.2.3
- Appearances vs. reality → §3.6, §30.2.5
- New vs. hereditary prince → §30.2.6
- Citizen militia → §30.2.7
- Cruelty well/ill-used → §30.2.8
- Liberality/parsimony → §30.2.9
- Ministers → §30.2.10
- Flatterers → §30.2.11, §36.5
- Compound growth → §15.7
- Factor tree → §17.7
- Forgetting → §10.5
- Memory (appearance vs. reality) → §6.7
- Power estimation → §14.6, §28.5
- Timing → §27.5
- Tempo/decisiveness → §35.4
- Terrain → §34.4
- Alliance instability → §33.4
- Intelligence → §32.4
- Person-reading → §19.6
- Ethics limit → §41.6
- Regimen → §38.5
- Trials → §39.4
- Sabotage → §40.5

APPENDIX F — NOTE ON SOURCES

This primer is a *mock* — a work of fan composition in the universe of the Dune novels of Frank Herbert (with the prequel and sequel novels by Brian Herbert and Kevin J. Anderson), and is not canonical. The Butlerian Jihad, the Mentat Schools, the Litany Against Fear, the figures of Thufir Hawat and Piter de Vries, the phrase "the Mentat computes. The Mentat does not choose," the naïve mind, the Mentat's Curse, sapho juice, melange, twisted Mentats, the Kwisatz Haderach, the six grades, Gilbertus Albans, Erasmus, Omnius, and the Mentat Mantra derive from Herbert's books (and the Herbert/Anderson prequels). The training regimen, tables, probability exercises, and pedagogical apparatus are invented for this document, though the underlying techniques (method of loci, spaced repetition, Fermi estimation, mental logarithms, base-rate reasoning, casting out nines) are real human methods of the pre-jihad world.

The attributed lectures are apocryphal. The arithmetic, however, is not.

The cross-references to Sun Tzu's Art of War and Machiavelli's The Prince are interpretive: the Schools have always studied both texts, and the mapping between their principles and Mentat doctrine is the Redactor's own, offered for the student's testing. The Mentat who disagrees with a mapping is encouraged to build a better one — disagreement is computation.

The attributed lectures are apocryphally attributed to Master Thufir Hawat, with later glosses by the Redactors of Salusa Secundus. The redaction is the third; the first two are lost. This is, perhaps, for the best.

Compiled for the amusement of a household. — Omnis, on the hub, 11,914 A.G. (or thereabouts).

END OF PRIMER